

Linguistic transparency as mediated accessibility

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Abstract

Linguists in several subfields call constructions **TRANSPARENT** (information remains available through a structure that could have hidden it) and mean different things by the label: number transparency in NP agreement, constructional compositionality (how a construction's parts contribute to its meaning), semantic transparency in compounds, and transparent free relatives. This paper treats those uses as one family of mediated accessibility relations (access to a property through an intervening element) and supplies a test for when the label is earning its keep. A configuration is transparent with respect to a property for a relation when the property, borne by one element inside the configuration, remains accessible to that relation despite another element whose own default behaviour would have concealed it. The distinctive move is a non-instance requirement: a case counts only where the mediator's default could have done the concealing, which excludes feature summation such as coordination, and with it the temptation to relabel familiar facts as access. The schema is a comparative concept in Haspelmath's sense rather than a claim that languages share a descriptive category.

The worked cases are English number-transparent quantificational nouns (nouns whose complements can control number agreement) and transparent free relatives. Spanish and Tsez extend the same outward-endpoint pattern; Aleut, Lardil, Kayardild, and Chamorro pressure-test the direction and distribution of the relation; and Algonquian/Kutenai obviation marks the boundary where an intervening head acquires a feature of its own, so the relation tracks that rather than the bearer's. Form–meaning accessibility, including constructional compositionality, compound transparency, and morphological priming, extends the schema beyond syntax, with cross-domain integration left as an open programme.

The result is a division of labour. The umbrella label predicts nothing on its own; particular transparency profiles do, and the paper says which ones, what each licenses, and what holds it in place. The pay-off for descriptive and theoretical grammar is a way to tell a transparency label that carries predictions from one that restates the facts it was coined for.

Keywords: mediated accessibility; transparency; agreement; comparative concepts; number transparency; transparent free relatives; projectibility

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29 AI USE. The large language models Claude Opus 4.7 & Opus 5 (1M context); OpenAI Codex 5.5; and GPT-
 30 5.6 Sol (pro) served as drafting and editing aids throughout the preparation of this paper. We're responsible
 31 for all theoretical claims, arguments, errors, and interpretive choices.

12 INTRODUCTION

33 The initial puzzle is that some grammatical relations can still access a property through a structure
 34 that appears, on the surface, to block or reverse that access. Two examples from Huddleston and
 35 Pullum's *Cambridge Grammar of the English Language* (CGEL, 2002) make the point visible (§18.2,
 36 p. 502).

- 37 (1) a. *A number of spots have/*has appeared.*
 38 b. *Heaps of money has/*have been spent.*

39 In the first example, the surface head *number* is singular, but agreement is plural because the *of*-
 40 complement *spots* is plural count. In the second, the surface head *heaps* is plural, but agreement is
 41 singular because the *of*-complement *money* is non-count. The construction lets the complement's
 42 number/count profile reach the verb, even though the surface head's own morphological number
 43 points the other way.

44 CGEL calls these NUMBER-TRANSPARENT QUANTIFICATIONAL NOUNS (QNs).¹ Core mem-
 45 bers include *lot*, *plenty*, *lots*, *bags*, *heaps*, *loads*, *oodles*, *stacks*, *remainder*, *rest*, *number*, and *couple*, with
 46 *majority* and *minority* at the boundary. They head NPs whose agreement override is, on CGEL's
 47 analysis, the default, unlike the optional override of ordinary collectives like *team* and *family*. The
 48 wider question is whether comparable *transparency* labels pick out real accessibility relations or only
 49 restate the facts they describe. The umbrella schema is:

Umbrella schema.

50 *X* is *transparent* with respect to property *P* for relation *R* when *P*, borne by one
 element within *X*, remains accessible to *R* despite another element of *X* whose own
 default behaviour would have concealed it.

51 The schema has three variables (*X*, *P*, *R*), two participant roles that fill out *X* (bearer and medi-
 52 ator), and two evidential controls (the diagnostic through which *R*'s behaviour is observed, and the
 53 concealing baseline the case is measured against). Keeping those apart is what stops a case study from
 54 drifting. Applied to (1a):

¹QUANTIFICATIONAL NOUN is CGEL's label, and the paper keeps it. In the pseudo-partitive literature the same nouns are QUANTIFIER NOUNS, one of five sub-types alongside measure, container, part, and collective nouns (Van Eynde & Kim, 2023, p. 271), following Keizer (2007, p. 109). Van Eynde and Kim print "collection" for this last type; the paper standardizes on "collective", following Kim (2026a).

<i>Configuration (X)</i>	<i>a number of spots</i> : a number-transparent quantificational noun plus its <i>of</i> -complement
<i>Property (P)</i>	plural count profile
<i>Bearer</i>	<i>spots</i> , the <i>of</i> -complement
<i>Mediator</i>	<i>number</i> , the morphologically singular head
<i>Relation (R)</i>	subject–verb agreement
<i>Diagnostic</i>	the number realized on <i>have</i>
<i>Concealing baseline</i>	head-driven agreement: <i>the number of spots has appeared</i>

Two of these are easy to run together and shouldn't be. The property is not its bearer: *P* is the plural count profile, not the word *spots*. And the relation is not the diagnostic that detects it: agreement is the relation, and verb morphology is how we see it. That second distinction does real work later, where evidence for accessibility exists without the relation having been specified (§3). The concealing baseline is what the schema's *would have concealed* clause points to, and §2.3 makes it the paper's falsification condition.

The substantive clause is *remains accessible through X*. A case fits only if *X* has internal structure whose default behaviour would have made *P* inaccessible to *R*. Coordination (§2.3) is the textbook exclusion. In *Bob and Jane are happy*, the plural agreement sums the conjuncts' features; no head's morphological number competes with them, so nothing was ever in a position to conceal *P*, and calling the construction *transparent* would relabel summation as access. Everyday senses of *transparent prose* or *transparent intentions* are excluded too; the schema's slots have no values to bind.

This non-instance requirement is the umbrella's distinctive contribution, and it's the part a working grammarian can use. It keeps mediated accessibility from collapsing into a relabeling of familiar facts: the schema earns its place not by naming a kind but by supplying a relational diagnostic that a per-construction analysis lacks, one that separates genuine mediation from compositional summation and from everyday CLARITY.

Passing that test is necessary and not sufficient. A label can pick out a genuine mediation relation and still tell us nothing we didn't already know, so the second question is which of these labels support prediction; call the ones that do PROJECTIBILITY PROFILES. Projectibility inherits the question Goodman (1955) raises, of which predicates support reliable inductive projection; Boyd (1991) supplies the stabilizer side with a maintenance-based realism. A profile has three components: an OBSERVABLE BASE (what identifies a candidate), the predictions that base licenses, and a STABILIZER (a source that keeps the cluster projectible). The QN class shows the shape concretely. The base is membership in CGEL's small class. The predictions reach well beyond the agreement override in (1): a noun in the class also shows a characteristic determiner frame (*a lot of it* and totality *the lot of it*, but not **some lot of it*) and restrictions on its complement (*a number of protesters* but not **a number of money*), among the behaviours §2 catalogues. The stabilizer is conventional lexical inheritance: the class is small and conventional, and membership carries the whole cluster as a package. So TRANSPARENCY, as an umbrella, isn't itself projectible: applied on its own the label projects nothing, and what projects is the particular profile it points to.

The paper therefore asks three questions. Which linguistic *transparency* labels bind a non-trivial *X-P-R* triple? Which triples support projection beyond the facts that motivated the label? What stabilizes the strongest profiles? The primary field is syntax, where the targets include agreement, syntactic-category licensing, complement selection, and behaviour under complement ellipsis. The

answer is strongest for number-transparent QNs and transparent free relatives.

What this buys descriptive and theoretical grammar, and typology, is a way to tell which of a field’s *transparency* labels carry predictions, and what those predictions are construction by construction. The philosophy of science here is instrumental. Goodman’s projectibility, Slater’s stable property clusters, and Khalidi’s causal networks are borrowed because they answer “does this label license inference?” precisely, and they’re put to work on grammatical descriptions rather than defended as metaphysics. A reader who wants only the grammatical results can take §2 and §4 and leave §5 to one side.

The English QN class illustrates the strongest case: once the class is recognized as small and conventional, membership predicts the whole cluster, not just the agreement override, reliably for core members and weakly at the edges (*majority*, *minority*, sense-split *couple*).

Transparent free relatives fill the slots differently. In *What appear to be trousers are on the chair*, *X* is the *what*-clause, *P* is the plural number profile borne by the nucleus *trousers*, the mediator is *what* (morphologically singular), and *R* is matrix subject–verb agreement. In attributional cases, a second relation appears: the matrix accesses the nucleus’s syntactic category through the *what*-clause.

A typological extension (§4) then tests the topology of the schema beyond English: Spanish and Tsez keep the outward endpoint shape, while the survey in Appendix B varies the source and direction of access (Aleut, Lardil, Kayardild, Chamorro) and marks the boundary where a feature is relayed onto a head rather than accessed directly (Algonquian/Kutenai obviation).

Projectibility is field-relative: a profile supports induction for a field’s purposes, not for every inquiry at once. *X* can be a lexical item, construction, compound, or derivationally complex word. *P* can be a number/count feature, constituent meaning, or stem representation. *R* can be agreement, interpretation, or masked priming. Accessibility itself can be class-like, as in core QNs, or gradient, as in compositionality ratings and priming magnitudes.

Our aim is constructive rather than exhaustive. The paper leaves aside most uses of TRANSPARENCY in linguistics, including extraction, islands, barriers, and phases; extraction-transparency is acknowledged in §2 but not developed, and Chamorro wh-agreement appears in §4 only as a topology comparison. It also leaves aside phonological or orthographic transparency, everyday talk of *transparent explanation*, and referential or *de re/de dicto* transparency, where the mediator is an intensional context rather than a form with internal structure and the schema’s *through X* clause has nothing to bind to.

MEDIATED ACCESSIBILITY is more specific than CLARITY and broader than COMPOSITIONALITY. Clarity is rhetorical: it doesn’t single out a mediator or relation. Compositionality is the limiting case of form–meaning transparency, where a mediator’s parts predict the whole. Number transparency is different: a QN’s morphological number doesn’t by itself predict the verb’s number, and under the construction it doesn’t have to.

Section 2 takes up morphosyntactic transparency, with number-transparent QNs and transparent free relatives as paradigms. Section 3 turns to form–meaning accessibility (constructional compositionality (Goldberg, 2006), rated compound transparency (Bell & Schäfer, 2016), and time-sensitive lexical processing (Heyer & Kornishova, 2018)), where the relation *R* shifts from agreement to interpretation and processing, and notes referential transparency as the schema’s boundary case; that family is developed in its own right in separate work.

Section 4 applies Haspelmath’s (2010) comparative-concept discipline to the schema’s cross-

linguistic ambitions and introduces the topology map, with Spanish and Tsez as the worked cases and the fuller topology survey in Appendix B. Section 5 names the stabilizers and treats Slater’s stable property cluster (SPC) framework (Slater, 2015) and Khalidi’s causal-network account (Khalidi, 2018) as complementary diagnostics.

MORPHOSYNTACTIC TRANSPARENCY

Morphosyntactic transparency is the case where X is a syntactic construction (or a lexical class within one) and R is a grammatical relation sensitive to a property borne inside X but not by its head. Feature matching between a non-head property and a co-occurring exponent is the paradigm, and agreement is the paradigm of that, but selection, category licensing, binding, and coordination all qualify, as §2.5 shows for transparent free relatives.² Subject-verb agreement is the textbook instance. The concrete contrasts are simple. In (1a), agreement tracks *spots* through singular *number*; in (14a), agreement tracks *trousers* through singular *what*.

This section carries the paper’s primary field-relative claim. The projected targets are syntactic and morphosyntactic: agreement, syntactic-category licensing, complement selection, behaviour under complement ellipsis, and distributional restrictions inside NP and fused-relative constructions. Nothing in the syntactic result depends on phrase-structure representation as such: the profiles require only observable relations among a mediator, an internal property, and the grammatical relation that tracks it, whether those relations are represented in constituency or dependency terms.

2.1 NUMBER-TRANSPARENT QUANTIFICATIONAL NOUNS

The paradigm again is *CGEL*’s two-direction contrast in (1): plural agreement despite singular *number*, and singular agreement despite plural *heaps*. Schema-wise (under the umbrella schema), X is the number-transparent quantificational noun + *of*-complement construction; P is the number/count profile of the *of*-complement; R is subject-verb agreement.

CGEL’s “simple agreement rule” is head-driven: “the verb agrees with a subject with the form of an NP whose person–number classification derives from its head noun” (§18.1, p. 499), as in (2).

- (2) a. *The nurse wants to see him.*
b. *The nurses want to see him.*

Number-transparent NPs override the rule. *CGEL* describes the override as obligatory in the central cases; rare singular-verb attestations with *number* are noted in footnote 73 to §18.2 (p. 502) and analyzed there as hypercorrections rather than as variant grammar. The contrast with the optional override of collective nouns (§2.2) is the operative point. The construction’s job is to let the *of*-complement’s number/count profile reach agreement past the head’s morphological number, which would otherwise have determined it.

²Extraction transparency (whether an island or barrier is opaque or transparent to long-distance dependency formation) is a morphosyntactic-accessibility relation under the schema (with R = dependency formation), and the associated question remains active. Transformational accounts posit empty elements (traces, copies) that mediate the connection between a fronted element and its base position; non-transformational dependency accounts (da Cunha & Gibson, 2025; Gibson, 2025) argue that no such mediating empty elements are needed and that direct head-to-dependent connections better fit acceptability and processing data. Both sides make schema-level claims about whether X contains invisible mediating structure. The centre of the argument here is the agreement-style cases.

The core inventory is small and well documented. *CGEL* identifies the principal members in example [57] (§3.3, p. 350): *lot*, *plenty* (singular forms; *lot* most often with *a*-determiner, *plenty* typically without a determiner); *lots*, *bags*, *heaps*, *loads*, *oodles*, *stacks* (plural forms, typically without a determiner); *remainder*, *rest* (singular forms typically with *the*-determiner); *number*, *couple*₂ (singular forms typically with *a*-determiner, taking only plural *of*-complements, partitive or pseudo-partitive). Of these, “the clear cases of number-transparent singular nouns are *lot*, *number*, and *couple*₂” (p. 503); *majority* and *minority* are borderline: plural override is “surely obligatory” in examples like *The majority of her friends are Irish*, but singular agreement is occasionally attested (§18.2, pp. 503–504). The determiner patterns above are typical, not absolute: *the lot of it* (in a totality sense) is grammatical alongside *a lot of it*, and similar sense-conditioned variations exist for other members.

Keizer’s (2007) study of English NP categorization and Van Eynde and Kim’s (2023) Head-driven Phrase Structure Grammar (HPSG) analysis of pseudo-partitives are the immediate background for treating these strings as structured binominal configurations rather than as a single loose family. Within the binominal family, Kim (2026a, pp. 98–103) distinguishes three sub-types by agreement head, illustrated in (3): measure nouns (*pound*), where the first noun controls agreement; quantity nouns (*lot*, *number*), where the second noun controls it; and collective nouns (*group*, *bunch*), which allow either.

- (3) a. Measure (N₁ controls): *One pound of tomatoes equals 90 calories.*
 b. Quantity (N₂ controls): *A lot of marbles are about to fall on the floor.*
 c. Collective (either): *A group of police cruisers is speeding across the bridge; a group of immigrants move in.*

The three example sentences are Kim’s, from COCA (Kim, 2026a, pp. 99–102). The quantity type is *CGEL*’s number-transparent class, and the measure type gives the schema its concealing variant: the same binominal string with agreement held by the first noun (§2.3).

Class membership predicts a cluster of behaviours beyond the agreement override. The relevant contrast is the standard PARTITIVE / PSEUDO-PARTITIVE distinction: partitive *of*-complements require a definite complement that picks out a subset of a definite set (*of the money*, *of those books*, *of these soldiers*), while pseudo-partitive *of*-complements express a quantity, measure, or collection relation with a bare or otherwise non-definite complement in the QN frame (*of money*, *of soldiers*, *of some paintings*) (Kim & Michaelis, 2020; Selkirk, 1977; Stickney, 2007; Van Eynde & Kim, 2023).³

Lot most often takes *a* as determiner with limited premodification (*A whole/huge lot of time has been wasted*; *CGEL*, §3.3, p. 350); *the lot of it* is also available in a totality sense, but other determiners are blocked (**some lot of it*). *Plenty* typically takes no determiner or modifier (*plenty of butter*, not generally *a remarkable plenty of butter*), and like the plural-only forms it strongly prefers the bare *of*-complement.⁴ The plural-only forms (*lots*, *bags*, *heaps*, *loads*, *oodles*, *stacks*, and *piles* beyond *CGEL*’s list) show the same pattern.

³The headedness issue is framework-internal. HPSG analyses treat the first noun as syntactic head in *A of the B* partitives, while *CGEL* retains the quantificational noun as head and treats agreement with the complement as override. Since the schema is framework-portable rather than neutral (§3), the paper doesn’t adjudicate the headedness choice; it records that the verdict is conditional on it.

⁴A COCA pilot (May 2026, extended July 2026) supports the preference but not a rate. The bare complement dominates for *plenty* and the plural-only forms, while *rest* and *remainder* prefer the determined one on matched complements ($\approx 97\%$ for *day*, *time*, *people*, *money*). No overall share is defensible: the two frames were sampled over different

- 203 (4) a. *They have oodles of money.*
 204 b. *Oodles of the money had already been spent.*

205 The contrast in (4) shows the plural-only pattern: the first is the characteristic use, the second a
 206 marked one. It is not a blocked one. A wildcard sweep of the determined frame attests it for every
 207 member of the subgroup, at 465 tokens across 374 distinct complements, where a probe over four
 208 hand-picked complements had returned 13 and two zeros (Appendix A.2). *Remainder* and *rest* are
 209 a different case again; *number* and *couple*₂ take plural obliques.

- 210 (5) a. *the remainder of the time*
 211 b. **the remainder of time* (partitive reading only; fine as “all time henceforth”)
 212 c. *the rest of society*
 213 d. *a number of the protesters*
 214 e. **a number of money*

215 The obliques in (5) are structured, but the lexical classification of *rest* is not settled by the corpus
 216 pattern. Under the semantic criterion adopted here, *rest* and *remainder* denote the complement sub-
 217 set of a presupposed whole rather than a quantity of it, so the pseudo-partitive construction, defined
 218 as one in which N₁ denotes “a quantity or amount” of what N₂ denotes (Van Eynde & Kim, 2023,
 219 p. 271), is unavailable. A competing analysis could treat bare *the rest of society* as pseudo-partitive by
 220 treating *rest* as a part or amount expression. The data establish the bounded-and-unique condition
 221 on the bare frame, but do not decide that lexical-semantic question; Table 1 therefore marks *rest* as
 222 unresolved while retaining the semantic analysis as the working description.

223 For the bare-complement reading, the whole must be uniquely identifiable and bounded, so that
 224 it has a proper part to take a remainder of. Generic collectives supply both unaided, which is why (5c)
 225 needs no determiner and is common: *the rest of society* alone has 442 COCA tokens, with *humanity*
 226 248, *nature* 78, *mankind* 77, *creation* 42, and *Congress* 21 behind it. An unbounded complement
 227 supplies neither. Bare *of time* offers no partition, which is why (5b) is starred on the relevant reading
 228 and why the string survives only on the unrelated duration reading, “all time henceforth”. That read-
 229 ing accounts for most of the corpus tokens of *the rest of time*, so those tokens are not evidence about
 230 the partitive/pseudo-partitive classification (Appendix A.2).

231 CGEL’s own minimal pair for *lot* has the complement controlling agreement in both directions
 232 (p. 349).

- 233 (6) a. *A lot of work was done.*
 234 b. *A lot of errors were made.*

235 The pair in (6) shows what number transparency is in the umbrella schema’s vocabulary. *X* (the
 236 construction) leaves *P* (the number/count profile of the *of*-complement) accessible to *R* (agreement).
 237 What *X* conceals from agreement is the morphological number of the head noun *lot*, *plenty*, *heaps*,

complement inventories, and frequency in any case supplies no denominator of occasions on which a partitive reading was intended. The partitive frame is attested for every member of both subgroups. Full counts and methodology in Appendix A.2; raw data in the supplementary material (see Data availability).

etc. The agreement-targeting feature is on the *of*-complement, not on the surface head. That's what makes *X* transparent with respect to *P*: *X* lets *P* reach the verb anyway.

The override is only visible where the head's number and the complement's diverge. (6b) shows it (singular *lot*, plural *errors*); (6a) doesn't, since head and complement both call for the singular. The discriminating cases are the crossed ones: a singular-form quantificational noun with a plural complement, or a plural-form one (*lots*, *heaps*) with a non-count complement. The matched cases are silent, not counterevidence. Transparency is therefore under-detected: a class can be transparent while most of its tokens look head-driven, so corpus frequency understates it.

2.2 COLLECTIVE NPs AS A TRANSITIONAL CASE

Singular collective nouns (*committee*, *jury*, *family*, *team*, *government*; also *clergy*, *intelligentsia*, *public*, etc.) allow optional plural override of head-driven agreement, as in *CGEL* [8] and [10] (§18.2, pp. 501–502).

- (7) a. *The committee has/have not yet come to a decision.*
- b. *The committee consists/*consist of two academic staff and three students.*
- c. *One committee, appointed last year, has/?have not yet met.*

The override in (7) is construal-sensitive: singular agreement tracks the unit construal, plural the member construal. Plural is unavailable with predicates that apply to the whole but not individual members, and is “of questionable acceptability” if the collective takes determiner *one* or *another* (p. 502).

Both collective NPs and number-transparent quantificational NPs override the head's morphological singular at agreement, and the override is possible because plural agreement requires only non-singular construal, not the exact enumeration that singular *a(n)* and low cardinals demand. That is why tight determiners degrade the override that (7a) allows while plural agreement survives it: the override operates at the loose end of the count cluster, not across it. Reynolds (2026a) develops that loose/tight dissociation at length.

The two cases differ in where the non-singular construal comes from. In collectives it is lexically encoded in the head, so the override is selective: it needs a predicate that licenses member-level construal, which is why whole-unit predicates block it. In number-transparent QNs it is imported from the *of*-complement, so the override applies regardless of predicate type. That difference matters more than it first appears, because only one of them satisfies the schema as stated. The schema requires one element within *X* to bear *P* and another to conceal it. In the QN case those are distinct expressions: the *of*-complement bears the number profile, the quantificational head conceals it. In *the committee have decided* they are not. The singular morphology and the member-level plurality are two representations of one lexical head, so there is no second element doing the concealing. Collectives are therefore a near case: a competition between morphological and semantic properties within a single sign, not access across a mediator. Broadening *element* to cover representational layers would accommodate them, at the cost of the non-instance test's grip, since almost any property could then be said to conceal another representation of itself. The transitional label in this subsection's title is meant in that strict sense.

Alongside *CGEL*'s override framing, the HPSG tradition encodes the same observation as index rather than concord agreement: concord tracks morphosyntactic features internal to the NP, while

index agreement can track the referential or semantic index available to the agreement target (Kim, 2004; Wechsler & Zlatić, 2003). Kim and Sells (2015, pp. 64–65) run the two levels on the cases at issue here: measure *four pounds was* is morphosyntactically plural but index-singular, while collective *this team have* is the mirror image, morphosyntactically singular with a plural index. That’s a different formal encoding of the same accessibility relation, and it’s consistent with the framework-portability of §3.

CGEL itself notes that “the division between collective and number-transparent nouns is by no means sharply drawn” (§18.2, p. 502). Some nouns straddle. *Couple* has two senses: *couple*₁ (a man and woman in a relationship) is a regular collective; *couple*₂ (‘approximately two’) is number-transparent. *Bunch* heads the same binominal *of*-frame as §2.1’s quantity nouns but is a collective noun in Kim’s (2026a, pp. 98–103) classification, the sub-type where either noun can control agreement; the straddling *CGEL* reports is what that classification predicts. *Bunch* is also the case where the profile framework does demotion work: *CGEL* claims it tilts collective for inanimate aggregates but transparent for groups of people, illustrated by the contrast in (8).

- (8) a. *A bunch of flowers was presented to the teacher.*
 b. *A bunch of hooligans were seen leaving the premises.*

CGEL treats the first as collective and the second as transparent (§18.2, p. 503). A pilot COCA check (May 2026) supports the animate half, where *bunch*-subjects are consistently plural, and finds no positive support for the inanimate half: inanimate *bunch* is rare in subject position, *CGEL*’s own example is unattested, and the few tokens that surface take plural rather than the predicted singular. Corpus absence of a constructed example isn’t ungrammaticality, so this doesn’t touch *CGEL*’s contrast as a claim about grammar; what it shows is that the inanimate-collective pattern doesn’t project as a corpus regularity. That localizes the projectibility claim rather than the grammatical one, and marking where a textbook contrast doesn’t project is what demotion conditions are for. Counts in Appendix A.4.

The umbrella schema accommodates both the paradigm and the transitional case, but they have different internal structure. Number-transparent NPs are class-like, with the override the default behaviour for class members. Collective NPs are gradient: the override probability depends on the head, the predicate, the determiner, the dialect register, and the construal the speaker has in mind. A theory of morphosyntactic transparency that ignored the difference between a class-based default override and a construal-driven optional one would lose the interesting structure of both cases.

2.3 COORDINATION AS A NON-INSTANCE

Coordination of singulars triggers plural agreement.

- (9) *Bob and Jane are happy.*

The ordinary coordination in (9) looks like a transparency case at first (the conjuncts’ number features percolate up to the coordinate construction and determine agreement), but the schema excludes it. The schema requires that *X could have* made *P* inaccessible. The coordinate construction does no such hiding: there’s no head whose morphological number competes with the conjuncts’ features for agreement.⁵ The plural agreement is compositional addition (1 + 1 = 2), not mediation.

⁵ This verdict presupposes a headless (or numberless-head) analysis: with no conjunct serving as a number-bearing

318 Some coordinate NPs take singular agreement.

319 (10) a. *Bacon and eggs is my favourite breakfast.*

320 b. *Wine and cheese is expected.*

321 c. *Fish and chips is expensive these days.*

322 Some of the same coordinate strings also take plural agreement.

323 (11) a. *Wine and cheese make the perfect party pair.*

324 b. *Fish and chips taste their best right after frying.*

325 The singular holistic cases in (10) and the plural counterparts in (11) leave the exclusion intact. In
 326 such cases agreement tracks a construal of the coordinate phrase: either a dish, package, event type, or
 327 conventional pairing, or a plural set of jointly mentioned items. The singular feature isn't a property
 328 of either conjunct that remains accessible through the coordinate construction; it's a construal of
 329 the coordinate phrase as a whole. Ordinary plural coordination and holistic singular coordination
 330 therefore differ in agreement semantics, but neither is transparent in the schema's sense. The former
 331 is compositional summation; the latter is whole-phrase unit construal.⁶ *CGEL* treats coordination-
 332 and-agreement under a separate heading (§18.4), not alongside the override constructions of §18.2.

333 The clean distinction across §2's cases: in collective plural agreement, internal member plural-
 334 ity becomes accessible through a singular head; in number-transparent QNs, complement number
 335 becomes accessible through a quantificational head; in unit-construal coordination, the whole co-
 336 ordinate phrase is construed as singular and no internal property is being made transparent. The first
 337 two are mediated accessibility; the third isn't.

338 The “*X* could have made *P* inaccessible” clause works as an independent diagnostic, not an ad
 339 hoc filter. The test asks whether the construction-type admits a concealing variant: is there a version
 340 of the same construction-type, attested in the language, in which *X*'s internal structure makes *P*
 341 inaccessible to *R*? For English NPs, head-driven agreement is the default and the number-transparent
 342 class is the marked exception within the same NP type. For coordinate constructions no such variant
 343 exists: the type admits no version in which conjunct features fail to reach agreement, so coordination
 344 has no concealing baseline to be measured against. The diagnostic distinguishes mediator-types from
 345 compositional-summation types: transparency is informative only where the same mediator type has
 346 opaque or partially opaque alternatives.

347 One condition has to be attached, or *same construction-type* becomes adjustable after the fact.
 348 Drawn narrowly enough, no transparent construction has an opaque sibling and every case is a non-
 349 instance; drawn broadly enough, almost anything does and the filter passes everything. So the con-
 350 cealing control must belong to a constructional family that is independently motivated, established

head, nothing conceals the conjuncts' summed features. The picture changes under an analysis on which one conjunct is the head. Closest-conjunct agreement (Bhatt & Walkow, 2013) is the case in point: where a verb tracks the nearest conjunct, that agreement conceals the resolved features and resolved agreement reaches them past it, so the same test can return an instance. As with the language-internal concealing baseline of §4, the verdict rides on the prior syntactic analysis.

⁶Cases like *One and one is two* are an arithmetic-expression construction, not a meal/package coordination: *and* functions close to an addition operator, and the subject denotes an operation, equation, or expression. Not transparency either.

on criteria that don't appeal to the transparency effect the test is assessing. English head-driven agreement qualifies: it is the unmarked pattern for NP subjects on grounds that have nothing to do with quantificational nouns, which is what makes the number-transparent class a marked exception *within* a family fixed elsewhere. Tsez class IV agreement with a clausal argument qualifies on the same footing. Where no such family can be identified without invoking the effect, the test returns no verdict rather than a negative one; the typological cases in §4 show why that control has to be fixed language-internally.

The condition also locates where the diagnostic is soft. Deciding what counts as the same construction-type is itself a language-internal analysis, so the test is crisp where the family is independently motivated and loose where it isn't. That is a real limitation on the paper's central instrument, not a detail about its edges.

A second non-instance fails on a different role, and the contrast is worth having because it answers the obvious deflationary reading of the QN case. In deferred or metonymic reference, agreement can track a referent the sentence never names.

- (12) a. *The hash browns at table nine are/*is getting cold.*
 b. *The hash browns at table nine is/*are getting angry.*

Kim's (2004, p. 1108) pair, which he credits to Nunberg, turns on which entity the subject is used to pick out: the food in the first, the customer who ordered it in the second. Warren (2006, p. 16) states the generalization, that metonymic subjects "need not agree as to number with their predicates" where metaphorical ones consistently do, and adds that a metonymic pronoun "sometimes agrees with the explicit and sometimes with the implicit element of the expression", the choice settled by context.

Filled into the frame of §1, the *bearer* cell is empty. The singular in (12b) is sourced from the designated customer, who is expressed nowhere in the subject NP, so there is no element within *X* bearing *P* and nothing for the relation to reach *through*. Coordination fails the test for want of a mediator; this fails for want of a bearer. Both are outside the schema, for different reasons.

Number transparency can be read as construal-tracking: the verb agrees with how the speaker conceives the subject, rather than with any constituent. Reference transfer shows what construal-tracking looks like on its own, and it looks different. It's context-dependent and reversible on one string, it isn't tied to a lexical class, and on Warren's account the pronoun may go either way. The QN override is none of these: near-categorical for core members, tied to a small conventional class, and construction-linked, as Appendix A.5's *a number of* versus *the number of* control shows on a single lexeme. Same surface phenomenon, different mechanism, and the frame's cells say which.

2.4 NUMBER-TRANSPARENT NPs AS A PROJECTIBILITY PROFILE

Section 1 set the diagnostic question: does identifying a candidate transparency profile let us predict behaviour? For number-transparent quantificational nouns the answer is yes, with the prediction running in two task settings: projection for the known members and class discovery for candidates. (The framework returns negatives too: the inanimate half of the *bunch* contrast fails to project, §2.2, and coordination never qualifies, §2.3.)

For known CGEL members (*lot*, *plenty*, *lots*, *bags*, *heaps*, *loads*, *oodles*, *stacks*, *remainder*, *rest*, *number*, *couple*₂), lexical identity is sufficient input: knowing a noun is on this list licenses projection to the rest of the profile.

For candidate new members, class discovery uses two diagnostics independently observable from any of the projected behaviours: quantificational semantics (an indefinite quantity rather than a precise count or kind) and high-frequency occurrence as head of an *of*-complement construction. A candidate that satisfies both is then tested against the projected cluster:

- near-obligatory agreement override in the central cases, singular- or plural-direction depending on subgroup;⁷
- characteristic determiner patterns by subgroup (*lot* most often with *a*, also with *the* in totality uses; *plenty* typically without a determiner; the plural-only forms typically without a determiner; *remainder*, *rest* typically with *the*; *number*, *couple₂* typically with *a*);
- partitive vs. pseudo-partitive *of*-complement availability;
- behaviour under complement ellipsis: the *of*-complement can be omitted while remaining understood, and it continues to determine the NP's number (*A lot was spent on travel* vs *A lot were arrested*; *CGEL* [56], p. 349);⁸
- count/non-count percolation of the *of*-complement to the whole NP (*lot* “allows the number of the oblique to percolate up to determine the number of the whole NP”; *CGEL*, p. 349), with consequences for selection by predicates that are count- or non-count-sensitive;
- restricted premodification (*a whole/huge lot* but not *a remarkable plenty*).

The class's sub-group structure makes the cluster's predictions concrete. Table 1 displays the per-noun pattern for the central diagnostics. The entries are *CGEL*-coded except in the two *of*-complement columns, where the COCA pilot has revised them: the determined frame is attested for every plural-only member, and *rest* and *remainder* admit a bare complement when it denotes uniquely. The semantic classification of *rest* remains open, and the per-noun corpus audit a fuller study would supply is the within-domain check that the falsification condition below leaves open.

Table 1. Number-transparent quantificational nouns: behavioural matrix (*CGEL*-based pilot).

Sub-group / Noun	Determiner	Partitive <i>of</i> N	Pseudo-partitive <i>of</i> N	Override direction
Singular, <i>a</i> -determined				
<i>lot</i>	<i>a</i> ; also <i>the</i> (totality)	✓ (<i>a lot of the money</i>)	✓ (<i>a lot of money</i>)	sg → pl with plural complement
<i>number</i>	<i>a</i>	✓ (plural oblique)	✓ (plural oblique)	sg → pl with plural complement
<i>couple₂</i>	<i>a</i>	✓ (plural oblique)	✓ (plural oblique)	sg → pl with plural complement

⁷A pilot COCA check (May 2026) supports the near-categorical pattern: predicted-direction agreement runs 98–100% across nine tested QN+complement combinations after KWIC filtering of checked counter-direction cells. *A lot of money* required parse-shift filtering (most raw plural-agreement hits are relative-clause artefacts; the full derivation is in Appendix A.3). The large *a lot of people is/was* counter-direction cell remains unfiltered, so the 98.0% row is conservative. The discriminating cells carry the evidence, and because filtering targeted counter-direction cells the figures corroborate the categorical pattern rather than estimate a rate. Full counts and methodology: Appendix A.3.

⁸These bare uses are not *CGEL*'s FUSED-HEAD construction, where the head fuses with a determiner or modifier (*Did you buy [some] yesterday?*; §9.1, p. 410). *CGEL* analyzes bare QN NPs as ordinary heads with an elliptical complement (pp. 412, 503). The fusion concept returns, differently applied, in the fused relatives of §2.5.

Sub-group / Noun	Determiner	Partitive <i>of</i> N	Pseudo-partitive <i>of</i> N	Override direction
Singular, usually without a determiner <i>plenty</i>	usually without a determiner	not typical	✓ (<i>plenty of money</i>)	sg → pl with plural complement
Plural, usually without a determiner <i>lots, bags, heaps, loads, oodles, stacks</i>	usually without a determiner	✓, dispreferred; attested for every member (§2.1)	✓	pl → sg with non-count complement
Singular, <i>the</i> -determined <i>remainder</i>	<i>the</i>	✓ (<i>the remainder of the time</i>); bare whole only if bounded and unique (<i>*the remainder of time</i>)	n/a: not a quantity noun	sg/pl (complement- controlled)
<i>rest</i>	<i>the</i>	✓ (<i>the rest of the time</i>); bare whole only if bounded and unique (<i>the rest of society</i>)	? (classification disputed)	sg/pl (complement- controlled)
Boundary <i>majority, minority</i>	<i>a</i>	✓	?	variable
<i>bunch</i>	<i>a</i>	✓	✓	variable

417 One sentence per central sub-group shows the override directions the table codes.

- 418 (13) a. *A number of spots have appeared.*
419 b. *Plenty of people were stranded.*
420 c. *Heaps of money has been spent.*
421 d. *The rest of the money was returned.*

422 (13a) and (13c) repeat (1): a singular *a*-determined noun driven plural by its complement, a
423 plural-only noun driven singular. (13b) shows determinerless *plenty* with the plural override; (13d)
424 shows the *the*-determined sub-group, where direction is complement-controlled (plural instead with
425 *the rest of the people*). The directions in all four match the COCA pilot's counts (Figure 1).

Override applies in the central cases for all class members; boundary cases show variable behaviour (§2.2). Complement-ellipsis behaviour and count/non-count percolation are predicted to apply across class members but aren't per-noun-audited here. Premodification is restricted across the class, though the restriction looks like one on modifier type rather than on premodification as such: degree and size modifiers are available (*a whole/huge lot*; *huge, enormous*, and *vast piles of money*, which with their kin carry 12 of the 15 premodified tokens in §2.5's COCA check), while evaluative ones are marginal (*a remarkable plenty*), and *plenty* takes minimal modification generally.

Each sub-group covaries: knowing a candidate noun's determiner pattern and partitive availability narrows the sub-group, which then licenses the override-direction prediction (and vice versa). One qualification on the partitive diagnostic. For *remainder*, the absence of a quantity reading follows from what kind of noun it is; for *rest*, the lexical-semantic classification remains unresolved. The diagnostic therefore carries less weight than the determiner and override behaviours (Appendix A.2). The boundary cases have weaker covariance, which is why §2.2's gradient framing applies to them.

Figure 1 visualizes the override-direction data from the COCA pilot. The probative cells are the discriminating ones, where the head's number and the complement's diverge; there the override runs in the predicted direction in every cell (plural with a singular-form quantificational noun and a plural complement, singular with a plural-form one and a non-count complement), and the counter-direction tokens that were checked largely resolve into parse-shift and hypercorrection artefacts, though the largest such cell was not filtered to the same depth. The matched cells confirm the predicted direction but cannot tell the override apart from head-driven agreement, so they corroborate without testing. The check is exploratory: counter-direction cells were hand-filtered by KWIC while predicted-direction cells were not scrubbed to the same depth, so the proportions read as attestation with documented exceptions rather than as estimated rates, and the small-*n* discriminating cells (*plenty of money*, *the rest of the people*) carry wide uncertainty. Referential *the number of* is a negative control: the same noun agrees head-driven under *the* and overrides only under quantificational *a* (Appendix A.5), so the override belongs to the construction, not the lexeme.

This two-task structure keeps the prediction genuinely projective for the new-member case: inputs are quantificational semantics and *of*-complement-head distribution; the cluster of agreement-override / determiner-frame / partitive / ellipsis / count-or-non-count / premodification behaviours is then testable on the candidate. Boundary cases like *majority* and *minority* (§2.2) and the dual-sense *couple*₁ (collective) versus *couple*₂ (number-transparent) show graded membership at the edges; the predictions hold in attenuated form, not as crisp class-conditional facts. What distinguishes prediction from redescription: the cluster supports inductive extension to *new* members (boundary cases like *majority/minority*; gradient cases like *bunch* in §2.2) that aren't built into the cluster's defining criteria.

The more demanding test is what the profile lets us predict beyond the in-grammar cluster. Two off-list projections are plausible candidates. *Diachronic*: members should enter the class through grammaticalization or sense-split rather than through arbitrary lexical addition, with recent boundary additions (*majority*, *minority*) and the sense-split in *couple* matching that pattern. *Cross-linguistic*: the comparative concept of mediated accessibility (§4) licenses asking whether languages with analogous partitive-quantificational configurations exhibit comparable cluster structure (French *beaucoup de*, Spanish *un montón de*); the descriptive categories needn't be equated for the cluster shape to be compared. None of these is tested in this paper. Their availability as predictions, rather than re-

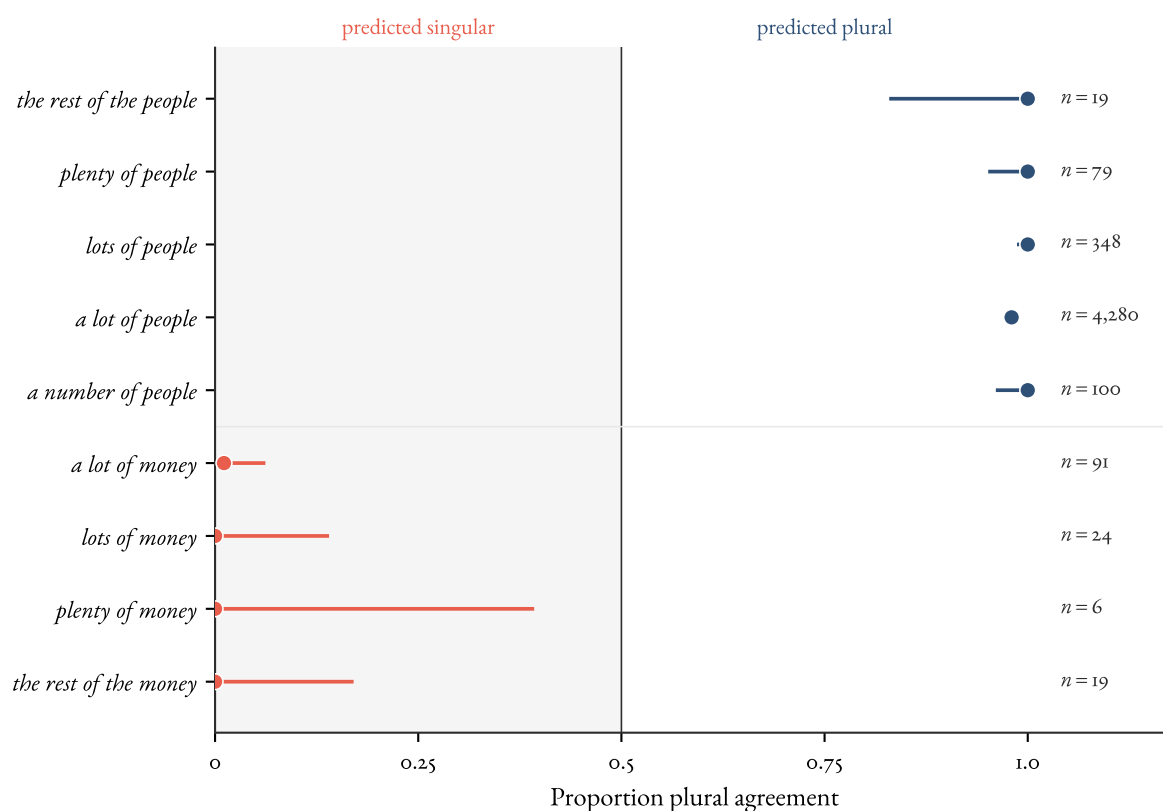


Figure 1: Agreement direction among adjacent finite-*be* tokens for nine number-transparent QN + complement combinations in COCA. Each row is a QN+complement; the x-axis is the proportion of plural-direction agreement (0 = all singular, 1 = all plural). Predicted-singular cases (coral) cluster at the left; predicted-plural cases (blue) cluster at the right. Horizontal bars are 95% Wilson binomial confidence intervals; n is the token count for that subject after KWIC filtering where available. The intervals cover token-sampling uncertainty only, not the larger uncertainties from query construction, parse contamination, and asymmetric hand-checking, so they should not be read as estimated grammatical rates. See footnote and the supplementary material. Data: COCA pilot, May 2026.

statements of the input criteria, is what supports treating the class as a projectibility profile in §5.1's stronger sense: a maintained-and-projectible cluster rather than just a distributional grouping. One candidate-member test has been run. *Piles* is not on CGEL's list and was admitted to the class here on two criteria only, quantity meaning and frequent *of*-complement distribution, neither of which is among the predicted behaviours. Of the predictions that follow, the determiner-frame one is confirmed and does unexpected work: *a piles of* returns no COCA hits, as the plural-only subgroup requires, and the determined frame *the piles of* N (340 tokens, 174 distinct complements) turns out to select the *literal* reading almost exclusively, its complements being things that can be heaped (*paper, books, debris, junk, rubble*), while the quantificational use concentrates on a few non-count complements. The determiner frame therefore separates the two senses of *piles*, which no prediction anticipated.

A follow-up sweep shows that separation isn't peculiar to *piles*. Across the plural-only subgroup the determined frame recruits the literal homonym wherever a member has one: *stacks of the ** is dominated by library stacks and smokestacks, *heaps of the ** by literal and colliery heaps, *bags* and *loads* by containers and cargo, and even *lots* carries a pharmaceutical batch sense (*lots of the drug, lots of the steroid*). *Oodles*, the one member with no literal homonym, returns quantificational hits exclusively. What looked like an accident of one candidate member is a subgroup regularity, and it yields a further prediction the profile can be held to: a candidate with a literal homonym should show the split, one without shouldn't.

The agreement prediction couldn't be tested at all. COCA has 91 tokens of *piles of money* and one followed by a finite form of *be*; adding COHA's two centuries brings the discriminating non-count cell to three tokens, one singular and two plural. Nothing there supports the override and three tokens cannot weigh against it either. A container-noun control (*sacks/boxes/crates of money*) returns nothing at all, and the left contexts of the 91 tokens locate the problem precisely: 41 have *piles of money* as the object of a verb (*make, spend, earn, amass, inherit*) and 13 as the complement of a preposition, so at least 59% sit in complement position, while the single token preceded by finite *be* is copular or existential, with agreement running from a different subject. Quantity expressions of this shape occur as complements, not as subjects of finite *be*. The same sparsity blocked the inanimate *bunch* contrast in §2.2.

So the override prediction, which is the profile's headline, is corpus-testable only for members that appear in subject position, and candidate-member discovery for this class will need acceptability judgments rather than frequency. That constrains how the profile can be tested rather than telling us anything about the profile itself. Protocol with predictions fixed in advance, full returns, and the corrections the wildcard queries forced on Appendix A.2 are in the supplementary material (see Data availability).

The cluster of behaviours listed above makes number-transparent quantificational nouns the strongest candidate PROJECTIBILITY PROFILE examined here: an observable base (lexical class membership for known members; quantificational semantics plus *of*-complement distribution for candidates) supporting projection to a structured set of targets, stabilized by conventional lexical inheritance. The profile is a candidate to pass Slater's (2015) stability-plus-projection diagnostic; the within-domain check beyond override direction is empirically open. The falsification condition is concrete: if a noun satisfying the input diagnostics failed to show the predicted cluster, the profile would fail. Boundary cases (*majority, minority, sense-split couple*) reflect the conventional inheritance that sta-

bilizes the profile. The count/non-count percolation behaviour the profile licenses participates in the broader count cluster that Reynolds (2026a) analyzes as Khalidian, with conventional individuation as a central causal node; §5.5 returns to it. Whether the profile earns the local-kind label as a downstream verdict is for §5.

2.5 TRANSPARENT FREE RELATIVES

A second morphosyntactic-transparency case fits the schema. Transparent free relatives (TFRs) are a sub-type of fused-relative construction (*CGEL*, ch. 12, §6, pp. 1068ff.) in which the matrix accesses properties of an embedded predicate (the NUCLEUS) through *what*: *what they call ecstatic*, *what we regard as essential*, *what seems to be a tourist*. FUSED is *CGEL*'s term for one word jointly realizing two functions: in a fused relative, "the head of an NP fuses with the relativised element in a relative clause" (p. 1073), so *what* in *what she wrote* is at once the NP's head and the relativized element. This paper uses TFRs only as a second profile case. The analysis followed here is Kim's construction-grammar one, developed across Kim (2011), Kim (2017), Kim (2026a, ch. 6.8), and its revision in Kim (2026b, §5), with the corpus material in Kim and Reynolds (2026). The verdict below is conditional on that analysis: rival treatments locate the mediator elsewhere, and where the mediator sits is exactly what the schema needs fixed, so a reader who prefers Wilder's (1999) parenthetical-with-backward-deletion account, van Riemsdijk's (2006) graft analysis, or Grosu's (2003) unified treatment should expect the slots to be filled differently: each locates the mediator, and so the access path, somewhere else.

- (14) a. *What appear to be trousers are on the chair.*
 b. *What they call essential workers are exhausted.*
 c. *He was utterly penniless and what they call dimwitted.*

For present purposes, the profile claim is enough. TFRs support two relations at different levels of the construction. One is agreement: matrix agreement accesses the nucleus's number profile through singular *what*. The other is matrix selection, whose accessible property is the nucleus's syntactic category; CATEGORY FLEXIBILITY is the outcome, that a non-NP nucleus can satisfy the matrix's requirement, as in (14c), where the nucleus is an AP. Predicative position admits both NPs and APs, so that example shows the nucleus's category rather than establishing that the construction as a whole occupies an AP-only slot; a category-discriminating environment would be needed for that, and the corpus material in Kim and Reynolds (2026) is where it belongs. Kim (2026b, pp. 36–37) treats the second as the marked sub-case of a wider SELECTION TRANSPARENCY, on which the matrix's selectional requirement is satisfied by the nucleus generally, NP nuclei included, as when a preposition selects the nucleus through the *what*-clause (*he speaks in what linguists call a Northern dialect*). The working-note data add *what I consider important* and *what he takes to be obvious*.

The observable base is constructional sub-type plus verb-class membership, and the examples instantiate both of the classes Kim (2026b, p. 40) distinguishes: the EVIDENTIAL type (*seem to be*, *appear to be*, *happen to be*; 14a), which takes NP nuclei only, and the ATTRIBUTIONAL type (*call*, *consider*, *describe as*, *regard as*, *take to be*; 14b, c), which licenses AP, AdvP, PP, and *-ing* nuclei as well. On the inheritance hierarchy Kim (2026b, p. 46) sets out, the parent licenses agreement, selection, binding, and coordination transparency across both classes, and the attributional daughter adds category flexibility. "Coordination transparency" there is Kim's name for a TFR diagnostic, that the construction conjoins as its nucleus's category; it doesn't reverse §2.3's verdict on ordinary coordinate NPs,

which concerns a different configuration. So the projected cluster is wider than the two relations this section uses, which is a point in the profile’s favour; the stabilizer is constructional inheritance. The verb-class corpus detail, AP/PP-nucleus stratification, and analysis of the attributional licence are deferred to Kim and Reynolds (2026).

2.6 BRIDGE

The cases in Section 2 share a relation: morphosyntactic matching mediated by a construction or lexical class. Section 3 shifts the relation, taking interpretation, rated transparency, and processing as *R* instead of agreement; the cross-linguistic reach of the schema, including the worked Spanish and Tsez cases and the topology survey, is Section 4’s topic, and the form–meaning arm is developed in separate work.

FORM–MEANING TRANSPARENCY

Section 2’s relation was agreement. The schema isn’t confined to it. Where *R* is interpretation, learning, or processing rather than agreement, the same question (does *P* remain accessible through *X* when *X* might have concealed it?) reaches a second family: the form–meaning transparency of constructions, compounds, and morphologically complex words. The relation is broader and the accessibility is gradient rather than class-like, but the slots bind as before. This section makes the case briefly; the form–meaning family is developed in its own right in separate work.

Three windows make the point, at three levels of structure: constructions, noun-noun compounds, and morphologically complex words. Take constructions first. Construction Grammar treats constructions as learned form–meaning pairings (Goldberg, 2006), and their compositional transparency runs a gradient: from fully compositional patterns, through partly schematic and idiomatically constrained constructions, to frozen idioms like *kick the bucket*, where the parts’ meanings no longer reach interpretation through the whole. The frozen idiom isn’t an exclusion but the trivial limit at the gradient’s opaque end (full compositionality, where a mediator’s parts predict the whole, is the transparent limit): the construction-type admits a concealing variant, so it sits on the scale rather than off it (contrast coordination, §2.3, whose type admits no concealing variant on the analysis adopted there). A plausible strengthening is that transparency is partly a *network* property, a construction’s compositional reading being accessible partly because related constructions in the CONSTRUCTION, the structured inventory of a language’s constructions, support it (Kay & Fillmore, 1999); that is a hypothesis, not a result.

The other two windows, at the compound and word levels, are quantitative and give the gradient empirical content. Bell and Schäfer (2016) model the rated transparency of English noun-noun compounds from corpus-statistical predictors (constituent frequency, semantic-relation expectedness, head productivity), so a compound’s parts are accessible to a degree that tracks its position in the network of compounds. Heyer and Kornishova (2018) find that rated transparency modulates masked morphological priming, but only at long stimulus-onset asynchronies: an early, semantically blind decomposition stage gives way to a later transparency-sensitive one. Their materials are deadjectival nominalizations in two languages: English *-ness* (*darkness* ‘state of being dark’) and Russian *-ost* (*blednost* ‘paleness’, from the adjective stem *bledn-* ‘pale’), both highly productive and mostly transparent but each shading into idiosyncrasy at the opaque edge (*business*, not ‘state of being busy’; *novost*, ‘a piece of news’ rather than ‘newness’). (Feldman et al. (2015) contest the serial reading; the dispute is live.)

These three windows are three within-domain profiles (construction-level, compound-level, word-level), distinct in items, methods, and stabilizers (constructional inheritance, usage entrenchment, processing architecture), and each licenses within-domain projection. Whether they're joint reflexes of a single underlying organization (a genuine cross-domain cluster) or three separate profiles is an open empirical question. Their composition hasn't been established (§5.3), and a within-items probe would test it; that probe and the family's fuller development are taken up in separate work. Here the windows do one job: they show the schema's *R* isn't fixed. A construction can be transparent for one relation and opaque for another. *Kick the bucket* is opaque for interpretation yet morphosyntactically regular, its verb agreeing with its subject in the ordinary way; the number-transparent NPs of §2 are constructions too, transparent for agreement whatever their interpretive compositionality. Accessibility isn't transitive across *R*-values. That *R*-relativity is built into the umbrella, and it's a feature: the schema asks whether *P* remains accessible *for the purposes of* a specified *R*, not in general. The schema is framework-portable rather than framework-neutral: CGEL, Sign-Based Construction Grammar, and Goldbergian CxG encode the override differently and the schema can be instantiated in any of them, but its verdict in a given case is conditional on an independently motivated analysis of the bearer, the mediator, and the path between them. That is a feature for the present purpose, since it lets the schema locate where frameworks disagree about transparency instead of standing above the disagreement.

TYPOLOGICAL TRANSPARENCY

This section is methodological. The previous sections worked through cases in English, with a Russian nominalization parallel in §3; the question here is how to move from those language-particular cases to any cross-linguistic claim. Haspelmath's (2010) descriptive-versus-comparative-concept distinction is the right discipline.

4.1 HASPELMATH'S DISTINCTION

Haspelmath (2010) argues that crosslinguistic comparison shouldn't assume universally available descriptive categories⁹ like ADJECTIVE, PASSIVE, ACCUSATIVE, or SUBJECT. Each language has its own descriptive categories, fixed by the language-particular criteria that make them work in that language (Haspelmath, 2010, p. 664). Comparative concepts, by contrast, are analyst-created tools for comparison; they aren't part of any language system and can only be more or less well suited to the comparative task, rather than right or wrong (Haspelmath, 2010, p. 665). They have to be built from universally applicable building blocks: conceptual-semantic notions, general formal notions, and other comparative concepts.

One pair makes the relation concrete. Haspelmath's comparative concept "dative case" is defined semantically, as a marker among whose functions is coding the recipient of a physical-transfer verb (Haspelmath, 2010, p. 666). Language-particular categories then match it or not independently of their traditional names: the Finnish Allative matches, while the Nivkh Dative-Accusative doesn't, despite the name, because it codes a causee rather than a recipient (Haspelmath, 2010, pp. 665–666). Categorial particularism, Haspelmath's preferred position, holds that descriptive categories are

⁹Throughout, *category* is used in Haspelmath's *descriptive category* sense: a language-particular grammatical grouping fixed by a language's own distributional criteria, and neutral on whether that grouping is projectible. This differs from the sense used elsewhere in this research programme, on which a *category* just is a projectible kind (Reynolds, 2026b); that kind-level notion is carried here by the *projectibility profile* (§5.1), not by *category*.

language-particular and that comparison runs through analyst-defined comparative concepts standing in many-to-many relations with them. The position it rejects, categorial universalism, assumes a substantial set of crosslinguistic descriptive categories from which languages select (Haspelmath, 2010, p. 663); Newmeyer (2007) is its canonical statement, taken up in §4.5.

The methodological consequence is straightforward. A typological generalization isn't a claim about a descriptive category instantiated in every language. It's a claim about how analyst-defined comparative concepts are realized in language-particular ways. That nominative case-marking is widespread doesn't entail that "nominative" picks out the same descriptive category in any two languages.

4.2 THE UMBRELLA SCHEMA AS A COMPARATIVE CONCEPT

The umbrella schema is a comparative concept, not a descriptive category. It's built from the materials Haspelmath (2010, p. 665) admits: *property* and *relation* are conceptual-semantic primitives, *form* is a general formal notion. Asking the schema's question of a language isn't claiming that the language has a category corresponding to TRANSPARENCY.

Accessible through is the contested slot, since accessibility has discipline-specific senses (discourse-pragmatic salience, psycholinguistic retrievability, formal-semantic substitution-licensing). The schema's sense is more abstract: *P* is accessible to *R* through *X* when *R* tracks *P* across *X* rather than tracking a property of *X* itself. The case studies operationalize that differently, by agreement targeting, constructional interpretability, rated transparency, or priming magnitude, but the comparative concept lives above the operationalizations.

The non-instance test (§2.3) is what keeps the concept honest, because its baseline is fixed language-internally. The test asks whether the language under study attests a same-type variant of *X* whose structure makes *P* inaccessible to *R*. For English number-transparent NPs that baseline is head-driven agreement, and the transparent class is the marked exception against it; the baseline is not exported. The procedure is identical in any language: identify *X*, *P*, and *R* by that language's own distributional criteria, then look for the concealing variant. Where it exists, the transparent case is informative; where the construction-type admits none, as with English coordination, the schema returns a non-instance. So *accessible through* imports no English-particular grammar. It inherits whatever default the local construction-type supplies.

So the bindings are language-particular while the question isn't. In English, *X* is the quantificational noun plus *of*-complement construction, a CGEL descriptive category; *P* is the complement's number/count profile; *R* is subject-verb agreement. In the worked cases below, Spanish partitive subjects and Tsez complement clauses fill the slots by different language-particular criteria. That's what makes the typological question tractable: not "do all languages have number-transparent constructions?", which would look for English-style NPs everywhere, but "which configurations in this language let a property remain accessible to a relation that the surface form might have hidden?" The answers are expected to differ, and the English, Spanish, and Tsez cases aren't instances of one descriptive category.

Two limits on what that buys. The claim at the comparative-concept level is structural, that the concept lives above the descriptive category whatever the language, rather than a demonstrated comparative result: the extensions on hand are the English–Russian nominalization parallel (§3) and the Spanish and Tsez cases below. And the form–meaning family (§3) isn't itself a comparative concept; the schema is, and the family is a set of language-particular phenomena and measurements compared through it.

4.3 CROSS-LINGUISTIC CASES AND TOPOLOGIES

Two worked languages let the question be asked outside English, one of them outside the agreement type English uses. Spanish has an analogous agreement-access pattern. In a partitive subject like *la mayoría de los manifestantes* ('the majority of the demonstrators'), the finite verb may agree in the plural with the complement or in the singular with the quantificational noun *mayoría*, as in (15).

- (15) a. *La mayoría de los manifestantes gritaban.*
 the majority of the demonstrators shout.IPFV.3PL
 'The majority of the demonstrators were shouting.'
 b. *La mayoría de los manifestantes gritaba.*
 the majority of the demonstrators shout.IPFV.3SG
 'The majority of the demonstrators was shouting.'

The plural (15a) is the more frequent option and the only one available under a plural predicative (Real Academia Española, n.d.). Both variants are attested on the one subject, much as English *majority* and *minority*, *mayoría*'s boundary cognates, take occasional singular agreement alongside the dominant plural override (§2.2). Spanish has its own literature on the alternation, which treats it as CONCORDANCIA AD SENSUM and reaches two conclusions that matter here. First, both options are genuine agreement rather than agreement displaced by government, so the plural isn't a species of SILEPSIS (San Julián Solana, 2018a). Second, the alternation isn't free across syntactic contexts, and singular agreement with the quantifier noun is compatible with a distributive reading of the clause (San Julián Solana, 2018a), which blocks the inference that notional plurality forces the override. The partitive and pseudo-partitive structures the alternation runs on are themselves heterogeneous in Spanish (San Julián Solana, 2018b), as they are in English (§2.1).

What travels to Spanish is the access pattern, not the English cluster: no Spanish determiner, complementation, ellipsis, or premodification profile has been shown here. And the concealment control is weaker than the English one, since the singular option with *mayoría* is dispreferred rather than blocked.

Tsez (Nakh-Daghestanian; head-final and ergative; about seven thousand speakers) gives a case in a different marking type. Tsez verbs prefix-agree in noun class with their absolutive argument, so the agreement is marked on the verbal head. A clausal argument is class IV, the class that includes clauses, so a matrix verb's default is class IV agreement with the complement clause as a unit; but when the embedded absolutive is a topic, the matrix verb can instead agree with it (Potsdam & Polinsky, 1999, p. 436).

- (16) a. *enir* [už-ā magalu b-āc'ru-ḷi] r-iyxo
 mother [boy-ERG bread.III.ABS III-ate-NMLZ] IV-knows
 'The mother knows the boy ate the bread.' (default: the matrix verb is class IV, agreeing with the clause)
 b. *enir* [už-ā magalu b-āc'ru-ḷi] b-iyxo
 mother [boy-ERG bread.III.ABS III-ate-NMLZ] III-knows
 'The mother knows the boy ate the bread.' (long-distance agreement: the matrix verb is class III, agreeing with embedded *magalu* 'bread')

In the schema's terms, *X* is the embedded clause, *P* is the noun class of the embedded absolutive, and *R* is matrix verb agreement. The concealing variant, and here genuinely the default, is the class IV agreement in (16a), attested in the same language and the same sentence, and (16b) is the transparent override. The baseline is not “the nearest head's own feature”, as with English number-transparent nouns, but “default class IV for a clausal argument”: a different default in a different language type. That is what the comparative concept predicts, the question constant and the language-particular answer not. Whether the override is analysed as covert movement or as long-distance agreement is a framework-internal question the schema brackets, as with extraction in §2; the fuller analysis is in Polinsky and Potsdam (2001).

Spanish and Tsez are both outward endpoint cases: a property associated with material inside *X* reaches an external agreement relation. The same schema maps several other accessibility topologies, provided each remains subject to the §2.3 concealing-baseline test: identify *X*, *P*, and *R*, then ask what same-language pattern would make the proposed access disappear. Appendix B sets out four of them with their falsification conditions (null-source subpart access in Aleut, inward dependent access in Lardil and Kayardild, path-distributed access in Chamorro, and the feature-relay boundary case in Kutenai and Algonquian obviation), along with a targeted check of head-marked systems.

Two results from that survey bear on the argument here. Aleut is a qualified positive: the controller of *R* can be recoverable as a subpart of a non-subject NP without being overt, though Merchant's analysis treats the unpronounced possessor as syntactically present *pro*, which keeps the case distinct from overt intact-NP transparency (Merchant, 2011, p. 195). And the head-marked candidates cluster around possessors, most of them analysed as externalized, by possessor raising, applicative mediation, or external possession, rather than as agreement with a dependent that stays NP-internal (Deal, 2017; Ritchie, 2016). So the diagnostic point is narrow: overt sensitivity to a possessor isn't enough, and the decisive question is whether an embedded dependent remains NP-internal while controlling clause-level morphology.

4.4 MAPPING TRANSPARENCY AND ACCESS TRANSPARENCY

There is an established typological programme under the name TRANSPARENCY, and it classifies the paper's paradigm cases as non-transparent. Setting the two side by side is the clearest available demonstration that TRANSPARENCY isn't one projectible kind.

Leufkens defines transparency as “a one-to-one relation between meaning and form”, treating every structure that violates such a relation as non-transparent (Leufkens, 2015, p. 2), and ranks 22 languages by how many non-transparent features their grammars show; the framework is extended to a larger sample and an implicational hierarchy in Hengeveld and Leufkens (2018). Opacity divides into four categories according to how the one-to-one relation fails (Leufkens, 2015, 16, §§4.1–4.4): redundancy, discontinuity, fusion, and form-based form. Agreement is the leading instance of the first. Where one unit of meaning is expressed by several forms, “one of the forms is not necessary; it is redundant, as it does not provide any additional information”, and clausal and phrasal agreement are catalogued accordingly (Leufkens, 2015, 18, §§4.1.3–4.1.4).

That is the opposite verdict from this paper's on the same configurations. Subject–verb agreement is the paradigm *R* for which a property remains accessible through a mediator; on the one-to-one conception it is redundant marking, and a language with more of it is less transparent. Both verdicts are correct within their own question. They are different questions: whether a language's levels map one-to-one, and whether a particular relation can reach a property across a mediator that

would otherwise conceal it. Call the first MAPPING TRANSPARENCY and the second ACCESS TRANSPARENCY.

English number-transparent QNs cross-classify. They are an access-transparency instance. Being agreement, they are also a mapping-transparency violation. A language could reduce its agreement, becoming more transparent in Leufkens's sense, and thereby lose the very relations that make access transparency visible. The divergence isn't a disagreement to be resolved by defining terms more carefully, because the two programmes individuate by different relations: levels of organisation in one, mediator-property-relation triples in the other. An umbrella that delivered opposite verdicts on one construction while both verdicts stood would not be a kind, and that is what §5 concludes on independent grounds.

Two further payoffs. Leufkens's inventory is organised, so it supplies a principled comparison class rather than another loose sense of the word, and the redundancy/discontinuity/fusion/form-based-form taxonomy is a checklist this paper's cases can be run against. And her finding that every language sampled has some non-transparent features, most often of the redundancy type (Leufkens, 2015), means agreement relations are common, so they are where a search for mediated access is most likely to find candidates. That says nothing about how often the search succeeds.

4.5 A POTENTIAL OBJECTION AND A REPLY

A reader might object that the umbrella schema looks like a universal descriptive category smuggled in as a comparative concept. The sharpest form of the objection targets the non-instance test: *accessible through* presupposes a concealing baseline, and for the English cases that baseline is head-driven agreement, which is independently the unmarked pattern, a language-particular fact. If the schema's deepest primitive is parasitic on an English-shaped default, then calling it universally applicable looks like categorial universalism in disguise.

The reply isn't that the schema is "just a question": a universal category can be framed as a question too, so that move wouldn't tell a comparative concept apart from a category in disguise. The reply is that the baseline the objection points to is fixed *language-internally*, which is how comparative concepts are meant to work. The concealing baseline is not assumed to be English head-driven agreement; it's whatever same-type concealing variant the language under study attests, by the procedure set out earlier in this section. So the schema doesn't import an English default; it inherits each language's own.

Three things follow that a universal descriptive category wouldn't deliver. The variables are bound by language-particular distributional criteria, not by a fixed cross-linguistic definition. The schema stands in many-to-many relations with descriptive categories: one language's category can fill *X*, *P*, or *R* under different bindings, and several categories can realize one binding. And it carries explicit falsification conditions: a candidate is a non-instance where the construction-type admits no concealing variant, and a profile claim is demoted where the predicted access fails to project, as in the *bunch* localization of §2.2, while the transparency attribution itself stands or falls on the grammatical facts. The honest residue is that the test is only as sharp as the prior individuation of construction-type: deciding what counts as the same type with and without concealment is itself a language-internal analysis, so the diagnostic is crisp where that individuation is independently motivated and softer where it isn't. The survey's boundary cases make the limitation visible: a candidate default that is not independently motivated cannot supply a confident positive. That the procedure declines to overgeneralize is a useful stress test, showing it doesn't mechanically convert

a superficially similar non-English case into a positive. It isn't evidence that the test is valid, since an underspecified test would also return indeterminacy. Haspelmath's (2010, p. 665) criterion for a good comparative concept, that it can be more or less well suited to the task of permitting crosslinguistic comparison, is exactly the standard the umbrella schema is meant to meet.

Newmeyer's specific argument deserves a direct response. His case is that typological generalizations must reference not just a universal concept but the specific form realizing it: negative particles and negative auxiliaries express the same concept, yet only the latter pattern with verbs in word-order typology, so a generalization stated over "negation" misses the regularity (Newmeyer, 2007, p. 136). The point is correct on its own terms. What follows from it is weaker than crosslinguistic *descriptive* categories, because the formal reference it demands enters at the *X* slot, and the slot and its values are different things. The slot is *typed* with general-formal notions in Haspelmath's (2010, p. 666) sense: construction, clause, NP, dependency domain. A value of the slot is a language-particular configuration fixed by that language's criteria, the number-transparent QN construction in English and the configurations the analyst identifies in each comparison language. The same holds for *P* and *R*: the slots are abstractly typed, while their values are particular number categories, grammatical relations, and agreement systems. Nothing here requires equating configurations token-for-token across languages. The concealing baseline that gives the non-instance test its grip is itself such a general-formal concept, not an imported universal category.

The metaphysical conclusion in §5, that the umbrella schema is a tool for locating projectibility profiles rather than itself a kind, is consistent with this. Three levels stay distinct: the comparative concept (mediated accessibility) lives at the cross-linguistic methodological level; descriptive categories (English number-transparent quantificational nouns, Spanish partitive subjects, Tsez long-distance agreement, and so on) live at the language-particular grammatical level; PROJECTIBILITY PROFILES are observable bases licensing non-trivial projection, stabilized by identifiable sources, realized within or across such descriptive categories. Kindhood is a downstream verdict on profiles that earn it. The paper doesn't collapse the levels.

Two limits on the position taken here. Haspelmath's distinction is itself contested and this paper doesn't adjudicate; it's adopted as the safer default, since it requires no commitment to descriptive-category universality, and a reader more comfortable with categorial universalism can take the schema as a working concept whose status is secondary to its methodological pay-off. And Croft's (2001) Radical Construction Grammar would route the same commitment through construction-similarity hierarchies rather than comparative concepts; the schema is meant to sit above any single construction-grammar variant, so that route is left open rather than argued against.

One consequence carries into §5: cross-linguistic projection at the profile level isn't entailed by cross-linguistic projection at the comparative-concept level. Asking the question everywhere doesn't make the answers travel.

PROJECTIBILITY PROFILES

Section 1 set the paper's central question: when does a transparency attribution support induction, and what stabilizes that induction? In Boydian terms, the question is field-relative: a profile is projectible for a field's explanatory and inductive tasks (Boyd, 1991, 1999). The primary field here is syntax. The strongest profiles support induction over agreement, syntactic-category licensing, complement selection, and related grammatical behaviour.

This section collects the answers and keeps the field-relative claims separate. The umbrella schema is a tool for locating projectibility profiles, not itself a profile, and the profiles the paper has examined are stabilized by heterogeneous structures.

5.1 THE UMBRELLA ISN'T A SINGLE PROFILE

Section 4 established that the umbrella schema is a comparative concept, an analyst's tool for asking the projectibility question of any language. The metaphysical question is what *follows* from posing the question across a heterogeneous family of cases. The umbrella's heterogeneity is the answer. The *X*-values across the family include lexical classes (number-transparent quantificational nouns), constructions (the ditransitive, partly-filled idioms), compounds (NN compounds), and derivationally complex words (*-ness* and *-ost*' nominalizations). The *P*-values include number/count features, constituent meanings and their relations, compositional contributions, and lexical-stem identity. The *R*-values include subject-verb agreement, compositional interpretation, and lexical access; masked priming and transparency ratings are the diagnostics through which the last two are observed, not relations in their own right. No single stabilizing source unifies these triples, and no single field's inductive base supports projection across them.

The umbrella identifies many candidate profiles, not one. A transparency label, on its own, licenses no projection: knowing only that something counts as *transparent*, that the schema's *X-P-R* slots are filled, lets us predict nothing substantive about it. The projectible content lives at the level of the individual profile, not at the level of the schema or the label. The schema's role is to make the projectibility question askable, case by case. That role isn't empty. The schema contributes the relational discipline of §2.3, the non-instance requirement that *X* could have made *P* inaccessible to *R*, and that requirement does analytical work no per-profile description supplies on its own: it excludes coordination as compositional summation (§2.3), places frozen idioms at the trivial limit of a real gradient rather than off it (§3), and marks where a *transparency* label picks out a genuine mediator rather than restating a distribution. The umbrella isn't a kind; its contribution is this relational discipline.

A useful general distinction follows from this heterogeneity. A *relational family* is a set of cases in which the schema's slots are all filled by something real, while a *projectibility profile* also requires some identifiable source of stability keeping the cluster projectible (a stabilizer in the sense of §5.4, not necessarily a causal mechanism). The family isn't a distributional class: its members share no surface distribution, which is precisely the point. The umbrella schema delimits such a family but not a profile: the schema picks out a real family of cases, but no single stabilizing source unifies them at the umbrella level. Each profile examined below has to be evaluated separately for whether it qualifies as projectible in this stronger sense. The negative-polarity-item literature offers a parallel pattern: NPIs share a surface licensing property but the licensing patterns Hoeksema (2012) catalogues aren't manifestations of one stabilizing source, so NPIs are a class but not a projectibility profile at the umbrella level (developed at length in (Reynolds, 2026b, ch. 7)).

5.2 TWO VERDICTS, KEPT APART

The word TRANSPARENCY has been doing three jobs, and separating them makes the framework operational. APPLICABILITY asks whether the case has an identifiable bearer, mediator, *P*, *R*, and concealing baseline at all. The RELATIONAL VERDICT asks, given applicability, how accessible *P* is to *R*. The PROFILE VERDICT asks whether identifying that relation licenses further projection. The

three come apart, and Table 4 records them separately for every case the paper has examined.

Table 3 assigns the §1 roles case by case. Setting them out together is what makes the difference between instantiating the schema and merely resembling its geometry visible: a blank in the bearer, mediator, or baseline column is why a case fails, and the last four rows are blank in the baseline column throughout.

Table 3. Roles filled, case by case. Em-dash marks a role the case doesn't supply.

Configuration	Property, and its bearer	Mediator	Relation (diagnostic)	Concealing baseline
QN + <i>of</i> -complement	number/count profile, borne by the complement	the QN, morphologically mismatched	agreement (verb form)	head-driven agreement
TFR, parent	number profile, borne by the nucleus	<i>what</i>	agreement (verb form)	ordinary fused relative
TFR, attributional daughter	syntactic category, borne by the nucleus	<i>what</i>	matrix selection (category flexibility)	evidential daughter, NP nuclei only
Collective NP	member plurality, borne by the head itself	— (same sign)	agreement (verb form)	—
Coordination	summed number, borne by no single conjunct	—	agreement (verb form)	—
Deferred reference	number, borne by the designatum	the subject NP	agreement (verb form)	—
Spanish partitive subject	number, borne by the complement	<i>mayoría</i>	agreement (verb form)	singular option, dispreferred not blocked
Tsez complement clause	noun class, borne by the embedded absolutive	the clause	matrix agreement (class prefix)	class IV, the clausal default
Frozen idiom	constituent meaning, borne by the parts	the construction	interpretation (paraphrase)	compositional sibling constructions
NN compound	constituent meaning, borne by the constituents	the compound	compositional interpretation (rated transparency)	opaque compounds
Derived word	stem identity, borne by the stem	the derived form	lexical access (priming magnitude)	opaque derivatives
Aleut (App. B)	person/number, borne by an unpronounced possessor	the possessed NP	anaphoric verb morphology plus subject case	overt-possessor alternant
Lardil/Kayardild (App. B)	clausal TAM, borne by the clause	the dependency domain	dependent-nominal morphology	—

Configuration	Property, and its bearer	Mediator	Relation (diagnostic)	Concealing baseline
Chamorro (App. B)	grammatical relation of the extractee	the extraction path	path morphology	—
Obviation (App. B)	possessor obviation, borne by the possessor	the possessed NP	— : the predicate tracks the head's own derived feature	—

888 Table 4. Relational and profile verdicts.

Case	Relational verdict	Profile verdict
Number-transparent QNs (§2.1)	Applicable, transparent	Strongest candidate; within-domain projection over known members, plus one candidate-member test (<i>piles</i>) with the determiner-frame prediction confirmed and the agreement prediction untestable for sparsity
Transparent free relatives (§2.5)	Applicable, transparent at parent (agreement) and daughter (selection)	Second syntactic candidate, conditional on Kim's constructional analysis
Collective NPs (§2.2)	Near case: bearer and mediator are layers of one sign, so not inter-element access	None claimed; the override is construal-sensitive rather than class-based
Ordinary coordination (§2.3)	Not applicable: no mediator, since nothing conceals the summed features	None
Deferred reference (§2.3)	Not applicable: no bearer within <i>X</i> , the property being sourced from the designatum	None
Spanish partitive subjects (§4.3)	Applicable, transparent	Not shown: the access pattern travels, the English cluster hasn't been tested
Tsez long-distance agreement (§4.3)	Applicable, transparent	None claimed
Frozen idioms (§3)	Applicable, opaque at or near degree zero	The gradient supports projection; the individual idiom doesn't

Case	Relational verdict	Profile verdict
NN compounds (§3)	Applicable, gradient; R = compositional interpretation, rated transparency the diagnostic	Within-domain, on corpus-statistical predictors
Derived words (§3)	Applicable, gradient; R = lexical access, priming magnitude the diagnostic	Within-domain, at long SOAs only

Two things follow. A case can be applicable and still score at the opaque end, which is why a frozen idiom sits on the gradient rather than off it, where coordination and deferred reference are off it entirely for want of a mediator and a bearer respectively. And a profile verdict can fail while the relational verdict stands: the thin inanimate *bunch* result (§2.2) weakens a proposed corpus-level regularity without touching CGEL's grammatical claim.

5.3 WHICH FIELDS THE PROFILES BELONG TO

Table 4 records the verdicts; what it doesn't show is which field each profile is projectible *for*, and that matters because projectibility is field-relative. The two syntactic profiles have observable bases a syntactician can apply: lexical class membership for the QNs, identified by quantificational semantics and *of*-complement distribution as well as by lexical identity, and constructional sub-type plus verb-class membership for the TFRs. Both project to grammatical behaviour, which is the field the paper claims.

The form–meaning profiles belong to neighbouring fields, and their bases are correspondingly different: corpus-statistical predictors for compounds, rated transparency for derived words. Each licenses within-domain projection. Whether the three compose into a cross-domain cluster on shared items is an open empirical question that the available evidence doesn't settle either way, and a within-items probe (corpus predictors, ratings, priming, interpretation accuracy on the same items), designed in separate work, would test it. The form–meaning family is an open research programme pursued elsewhere, not a delivered result of this paper.

5.4 STABILIZING STRUCTURES

Different structures stabilize the profiles, and the paper's R -values across §§2–3 call for a small inventory of stabilizing sources rather than a single mechanism.

LEXICAL CONVENTION stabilizes the number-transparent nouns as a class. Lexical inheritance here means that speakers acquire not isolated words but item-specific constructional slots: *a lot of X*, *plenty of X*, *the rest of X*, *a number of X*, with their associated agreement and complement-selection behaviour. The acquisition target is the slot, not just the lexeme. Speakers acquire a small, relatively stable inventory of these slots; new members enter at the boundary (*majority*, *minority*) or by sense-split (*couple₂*). Class identity itself isn't held by Boyd-style homeostatic feedback; it has the stability characteristic of any small, conventionally established grammatical class, and the override it licenses is a property of that construction, corroborated by the negative control in Appendix A.5. The count/non-count percolation behaviour the class licenses connects to broader patterns in English countability; (Reynolds, 2026a) develops a Khalidian causal-cluster analysis of those patterns

with conventional individuation as a central node. The transparency paper draws on that analysis but doesn't depend on its stronger claims; the within-NP percolation can be observed and predicted independently.

CONSTRUCTIONAL INHERITANCE stabilizes the TFR daughter profile (§2.5) and the network-relative transparency hypothesis of §3. In TFRs, the parent and attributional daughter support different access conditions. In the form–meaning family, a construction's compositional accessibility depends partly on the inheritance hierarchy it sits in; constructions whose inherited supercategories are well-attested support stronger compositional projection than orphaned constructions.

USAGE-BASED ENTRENCHMENT stabilizes the corpus-statistical predictors of compound transparency (§3). Bell and Schäfer's frequency, productivity, and semantic-relation expectedness predictors are corpus-derivable measurements of usage entrenchment; the rated-transparency target is a steady-state property whose stability is plausibly a reflex of that entrenchment.

PROCESSING ARCHITECTURE stabilizes the time-course modulation of priming (§3). Heyer and Kornishova's SOA-dependent transparency effect is a processing-architecture observation: the time-course of morphological decomposition makes morpho-orthographic information available before morpho-semantic information, and the resulting priming pattern depends on representational properties that the processing architecture exposes only at certain stages.

The list isn't closed. Other linguistic profiles may be stabilized by combinations of these or by sources the paper doesn't invoke (information-structural pressures, learnability constraints, normative/pedagogical regularization). The point is that “what stabilizes this profile?” is the right empirical question, and the answer is heterogeneous.

5.5 SPC AND KHALIDI AS DIAGNOSTICS

SPC and Khalidi are complementary diagnostics rather than nested ones. Slater asks whether the cluster is stable and projection-supporting and is neutral on *what* stabilizes it. Khalidi asks whether the stabilizing structure is itself causal-hierarchical, with named candidate core properties whose causal action generates derivative properties. A profile can satisfy SPC without satisfying Khalidi, and (in principle) the reverse. The PROJECTIBILITY PROFILE is the unit of analysis; the two diagnostics ask different questions of it.

Slater's (2015) stable property cluster (SPC) account gives the stability-plus-projection diagnostic. On it, a category qualifies when its cluster of properties is, in Slater's phrase, “sufficiently stably coinstantiated to accommodate the inferential and explanatory uses” a science puts the category to, regardless of *what* maintains the stability. The affirmative content is that the stability has to be CLIQUISH: possessing some of the cluster's properties reliably (if imperfectly) indicates the rest, so that observing a few licenses projection to the others. Number-transparent quantificational nouns are the strongest candidate to satisfy SPC examined here. Concretely, cliquishness means that a noun showing a few of the class's properties (the *a lot of* determiner frame, a strong preference for the bare *of*-complement, the agreement override onto its complement) is a good bet to show the rest; that is the sub-group covariance §2.4 documents, where determiner-pattern, partitive availability, and override-direction co-vary by sub-group. The maintenance is at least conventional/lexical for class identity (§5.4). The within-domain audit beyond override direction is empirically open: the cliquishness claim is licensed by that covariance, not yet by an end-to-end corpus or experimental study.

TFRs are a second syntactic candidate. The parent/daughter split and the verb-class restriction

form a projected cluster stabilized by constructional inheritance, though the present evidence is a working corpus check rather than a completed profile audit. Each form–meaning window (§3) is a candidate to satisfy SPC within its own field, pending the within-domain audits. Whether the family as a whole satisfies SPC at the cross-domain level is the open empirical question §5.3 flags.

Khalidi (2018) applies where the stabilizing structure is ordered and causal-hierarchical, with named core properties whose causal action generates derivative properties via identifiable pathways. QN class identity isn’t such a case: it’s conventionally stabilized without a separate ordered causal hierarchy, so SPC, not Khalidi, is the diagnostic that fits it. One level out, the count cluster the QN’s *of*-complement feeds (*a lot of money* non-count, *a lot of tourists* count) has been argued to be causal-hierarchical, with conventional individuation as a candidate core property (Reynolds, 2026a). That analysis belongs to the countability paper and isn’t needed here: the within-NP percolation the QN profile relies on can be observed and predicted under SPC alone. Khalidi names the vocabulary for asking whether a sub-profile has internal causal ordering; it isn’t a result this paper delivers for the QN case.

For the form–meaning family of §3, Khalidi is acknowledged as a possible refinement *if* corpus predictors, rated transparency, priming magnitude, and interpretation accuracy align on shared items as joint reflexes of underlying representational structure. These four measures aren’t a temporal chain (masked priming probes early lexical access before explicit interpretation, and ratings are a late meta-linguistic judgement) but candidate joint consequences of a single underlying organization. The within-items evidence doesn’t yet exist; separate work sketches a within-items probe design. Until that probe is run, Khalidi remains a possible refinement for the form–meaning family rather than a delivered analysis.

Kindhood, on this picture, is the downstream verdict. A profile that scores well on the SPC diagnostic, and especially one that also scores well on the Khalidi diagnostic, has earned the kind label. A profile that scores modestly is a candidate. A profile that fails the SPC diagnostic isn’t a kind, even if it remains a useful descriptive category. Failing Khalidi’s is not disqualifying, since the two aren’t nested: a conventionally stabilized cluster can be stably coinstantiated without its stabilizers forming an ordered causal hierarchy, which is exactly the QN case. What supports projection is the stabilizing structure; the diagnostics assess whether it does so well enough, and the kind label follows from that assessment.

The apparatus matters in two places. SPC’s maintenance-neutrality lets the paper put conventional, entrenched, architectural, and formal-semantic stabilizers on the same footing, where Boyd’s homeostasis would force one mechanism (a poor fit for lexical convention). Khalidi’s causal-hierarchy diagnostic gives the paper a vocabulary for asking whether sub-profiles have internal ordering: the form–meaning family’s possible cross-modal cluster is the candidate case.

5.6 THE RESULTING CLAIM

Linguistic TRANSPARENCY isn’t a kind, and the umbrella schema doesn’t aspire to be one. Applying the label predicts nothing on its own: that something is *transparent* says the schema’s slots are filled, not what the filling lets you infer, so what supports induction is the specific profile a transparency claim points to, not the word. The schema is a diagnostic tool for locating field-relative projectibility profiles: cases where an observable base licenses non-trivial projection to a target, stabilized by some identifiable source. Different transparency profiles are stabilized differently (lexical convention, constructional inheritance, usage entrenchment, processing architecture, formal-semantic en-

vironment), and each profile’s projectibility is what makes the transparency attribution scientifically useful, not the profile’s classification within any single ontology of kinds.

Three levels stay distinct, per §4:

- COMPARATIVE CONCEPT: mediated accessibility, the cross-linguistic methodological tool defined by the schema of §1.
- DESCRIPTIVE CATEGORY: language-particular grammatical content. English number-transparent quantificational nouns are a CGEL descriptive category; English NN compounds and *-ness* nominalizations are descriptive categories defined by language-particular criteria; Spanish partitive subjects and Tsez long-distance agreement are language-particular descriptive categories; and so on.
- PROJECTIBILITY PROFILE: an observable base that licenses non-trivial projection to a target, stabilized by some identifiable source, realized within or across language-particular descriptive categories. This is the projectible profile of §5.1: a maintained-and-projectible cluster, contrasted with a *class* (mere distributional grouping). Profiles that score well on the SPC and (where appropriate) Khalidi diagnostics earn the local-kind label as a downstream verdict.

The contribution here is to apply the projectibility-first move to a *relational* concept. TRANSPARENCY is the relation *X*-mediates-*P*-for-*R*, not an entity. Treating a relation this way needs something a property-kind analysis doesn’t: a way to tell genuine mediation from cases where no property is being made accessible at all. The non-instance requirement of §2.3 supplies it, which is why the projectibility-first move transfers to a relational concept instead of dissolving into a list of co-occurrences. Asking *when does this attribution support induction, and what stabilizes the induction?* works for relations as it does for the umbrella concepts *language* and *grammatical category* addressed in related work. The paper asks that naturalization question of transparency too, and answers it locally, profile by profile, rather than at the umbrella level.

CONCLUSION

Linguistic transparency is a family of mediated accessibility relations, not a single syntactic property. The umbrella schema (*X is transparent with respect to P for R when P, borne by one element within X, remains accessible to R despite another element of X that would have concealed it*) is a comparative concept: an analyst’s question, applicable to any language, asking which configurations let a property remain available to a grammatical or interpretive relation across a mediator.

Its distinctive contribution is a non-instance requirement: a configuration qualifies only when the mediator’s own default behaviour could have concealed *P* from *R*. That filter separates genuine mediation from compositional summation and from everyday CLARITY, keeping the schema from relabeling familiar facts. It doesn’t claim that any language has a descriptive category called TRANSPARENCY.

The main field-relative result is syntactic. Number-transparent quantificational nouns are the strongest candidate examined: a small lexical class whose membership predicts a cluster of correlated behaviours (agreement override, determiner patterns, partitive availability, complement-ellipsis behaviour, count/non-count percolation, restricted premodification), stabilized by conventional lexical inheritance; the override claim rests on the COCA pilot of Appendix A.3, and partitive *availability* is corpus-attested for every class member, though the pilot fixes no rate and how strong the preference is rests on descriptive analysis and judgment (Appendix A.2). Transparent free relatives supply a

second syntactic profile, with constructional sub-type and verb-class membership predicting parent-level agreement transparency and the attributional daughter's category flexibility.

The form–meaning accessibility family (constructional compositionality, rated compound transparency, masked-priming morphological transparency) gives three adjacent within-domain profiles whose cross-domain integration on shared items remains an open research programme, developed in separate work that designs the within-items probe to settle it. Coordination is excluded as a non-instance, and so is referential transparency, where the mediator is a context rather than a form.

The typological cases widen the schema beyond outward escape from an internal property. English QNs, TFRs, Spanish partitives, and Tsez long-distance agreement instantiate outward endpoint access; the further topologies collected in Appendix B add three more, with Aleut a qualified positive for null-source subpart access, and inward dependent access and path-distributed access as candidate topologies illustrated by Lardil and Kayardild and by Chamorro wh-agreement. Algonquian/Kutenai obviation marks the negative boundary, on a criterion Appendix B states: where an intervening head acquires a feature of its own and the predicate tracks that, the bearer has been replaced rather than reached. The criterion is analysis-dependent, and the verdict travels with the analysis.

Those additions widen the topology without promoting every case to a worked projectibility profile. Aleut matters most for the head-marking question §4 raises (Appendix B), but Merchant's analysis keeps it distinct from overt intact-NP transparency. Lardil, Kayardild, and Chamorro pressure-test directionality and distribution of *R* while leaving QNs and TFRs as the paper's core syntactic profiles.

Other multi-sense umbrella concepts (*language*, *grammatical category*) admit the same projectibility-first move: the umbrella isn't itself a kind, but specific senses can be projectible profiles. Reynolds (2026b, ch. 7) develops the structural analogue for negative-polarity items at length.

A transparency attribution is scientifically useful when it identifies a field-relative projectibility profile: an observable base that licenses non-trivial projection to a target, stabilized by an identifiable source. Some profiles support strong projection; others weak; some none. Kindhood is the downstream verdict on profiles that earn it, not the master question of the analysis.

107A NUMBER-TRANSPARENT NP EMPIRICAL APPARATUS

1078 This appendix collects the COCA pilot data behind the §2 claims about the number-transparent quantificational noun class. The body of the paper
 1079 carries the qualitative claims; the audit trail lives here. It covers the two cluster behaviours with direct corpus data, the partitive/pseudo-partitive contrast
 1080 (§A.2) and the override direction (§A.3), plus the *bunch* localization (§A.4) and the negative control (§A.5); the remaining cluster behaviours are CGEL-
 1081 coded predictions, not separately corpus-audited. Full machine-readable data and KWIC spot-checks are in the supplementary material (see Data
 1082 availability).

1083 A.1 METHODOLOGY

1084 Queries were issued through the local Playwright wrapper at tools/english-corpora/bin/ecorg.mjs (May 2026). Each query is a surface string and
 1085 the counts are raw COCA frequencies, filtered for head-versus-modifier use, idiom contamination, and sense only where noted. Two confounds recur
 1086 and both inflate counts in the counter-predicted direction: a MODIFIER CONFOUND, where N in X of N is a modifier inside a larger NP (*lots of day-care*
 1087 for *lots of day*), and a PARSE-SHIFT, where the QN sits in a relative clause and the visible verb agrees with a different head (of 13 raw plural hits for *a lot*
 1088 of *money are*, 10 are parse-shifted, 1 is irrealis, 1 is contracted *bas*, and 1 is a genuine override).

1089 Filtering was applied asymmetrically: counter-predicted cells were checked by hand more thoroughly than predicted ones, because the wrapper's
 1090 KWIC mode was returning a Playwright error on every scripted call and checks had to be done manually and selectively. The counts below should
 1091 therefore be read as attestation with a known bias in the paper's favour, not as an estimated grammatical rate. The full query list, the raw returns, and
 1092 the per-cell coding decisions are in the supplementary material (see Data availability).

1093 A.2 PARTITIVE VS. PSEUDO-PARTITIVE OF-COMPLEMENTS

1094 The contrast: PARTITIVE *of*-complements have definite complements (*of the money, of those books*); PSEUDO-PARTITIVE *of*-complements have bare or
 1095 otherwise non-definite complements in the QN frame (*of money, of soldiers, of some paintings*).

1096 A.2.1 Plural-only forms (*lots, bags, heaps, loads, oodles, stacks, piles*)

1097 Pseudo-partitive (X of N): *lots of people* 3,878; *lots of money* 1,376; *lots of time* 703; *loads of money* 109; *piles of money* 91; *loads of people* 90; *loads of*
 1098 *time* 34; *heaps of money* 15; *heaps of people* 7; plus single-digit hits for several other combinations. Total: 6,311. KWIC checks of the single-digit cells
 1099 show the expected mixture: *lots of day* is modifier use (*day-to-day, day hikes, day trips*); *heaps of time* and *piles of time* are genuine pseudo-partitive
 1100 non-count-complement uses; *piles of people* is literal piling rather than the quantificational use. *Piles* is not among CGEL's [57] core forms but patterns
 1101 with the plural-only class, so it is included in the totals here.

1102 Partitive (X of the N), hand-picked complements: *lots of the time* 7; *lots of the people* 5; *loads of the day* 1, with 0 for *piles* and *heaps*. Total: 13.
 1103 Those figures are superseded. A wildcard sweep of the same frame (X of the *, one query per form, July 2026) returns *lots* 313 tokens across 235 distinct
 1104 complements; *stacks* 47/42; *loads* 38/33; *bags* 38/35; *heaps* 14/14; *piles* 12/12; *oodles* 3/3. Total: 465 across 374 distinct complements, some thirty-six
 1105 times the hand-picked count. The partitive frame is attested for every member of the subgroup, and the zeros in the earlier probe were artefacts of
 1106 testing four complements.

1107 What the sweep does show is a sense split, and it generalizes the *piles* result of §2.5 across the whole subgroup. Wherever a member has a literal
 1108 homonym, the determined frame recruits it: *stacks* is dominated by library stacks and smokestacks (*the library* 4, *the feed* 2, *the mill* 2, with singletons
 1109 in *the plant, the power, the coal, the oil, the factories*); *heaps* by literal heaps and colliery spoil (*the dead, the brush, the dross, the Rhondda*); *bags* and *loads*
 1110 by containers and cargo (*the chips, the greens, the soil, the biosolids, the shellfish*). Even *lots* carries a pharmaceutical batch sense (*the drug* 3, *the steroid* 2,
 1111 *the plasma, the methylprednisolone*). *Oodles*, the one member with no literal homonym, returns three hits and all three are quantificational (*the cute, the*
 1112 *sheepie, the stuff*). The clearly quantificational partitives concentrate on complements naming an already-established set: *the same* (*lots* 11, *loads* 2), *the*
 1113 *stuff* (*lots* 3, *bags* 3, *loads* 3, *heaps* 1, *oodles* 1), *the latter, the usual, the other*, and definite human plurals (*the women, the people, the kids, the guests, the*
 1114 *members, the children*). Sense assignment here is read off the complement types, not off inspected concordance lines, so it identifies where the literal
 1115 reading concentrates without measuring its share.

1116 Indefinite-NP complement (X of a N): *lots of a day* 0; *piles of a day* 0; *heaps of a day* 0; *loads of a day* 0; *lots/piles/heaps/loads of an afternoon* 0.
 1117 Total: 0. That's a null result over eight specific strings, and given what the wildcard did to the definite cells it shouldn't be read as showing the frame
 1118 unavailable. Testing it properly would take its own sweep.

1119 No partitive share is reported. The wildcard numerator and the hand-picked pseudo-partitive denominator come from different complement
 1120 inventories, so dividing one by the other would measure the sampling rather than the language. The defensible statement is qualitative: both frames
 1121 are available across the subgroup, and the bare complement is by a wide margin the characteristic one.

1122 A.2.2 *Plenty*

1123 Pseudo-partitive: *plenty of time* 4,017; *plenty of people* 1,563; *plenty of money* 668; *plenty of day* 4 (likely modifier). Total: 6,252.

1124 Partitive: the two hand-picked complements return *plenty of the time* 1 and *plenty of the people* 1. The wildcard *plenty of the ** returns 169 tokens
 1125 across 140 distinct complements, led by *the latter* 7, *the same* 6, *the usual* 4, *the things* 4, *the tools* 3, *the real* 3, *the old* 3. *Plenty* has no literal homonym,
 1126 so the sense confound affecting *stacks, heaps, bags, loads*, and *piles* doesn't arise, and the count stands as quantificational partitive use. The near-absence
 1127 reported from the two-string probe was an artefact of the two strings. No ratio, for the reason given above.

1128 A.2.3 *Rest, remainder*

1129 Under the semantic criterion adopted in the body, *rest* and *remainder* are not quantity nouns, and that changes what the two frames mean for them.
 1130 Van Eynde and Kim (2023, p. 271) define a pseudo-partitive as [N1 of N2] in which "N1 denotes a quantity or amount of whatever it is that N2
 1131 denotes", and treat the bare N2 as "characteristic" of the construction, the property by which it differs from a genuine partitive, "in which N2 is
 1132 introduced by a definite determiner". For *lot* and the plural-only forms the semantic and syntactic criteria coincide. For *rest*, they may come apart: a
 1133 competing analysis could treat bare *the rest of society* as pseudo-partitive by treating *rest* as a part or amount expression. *The remainder of time* remains
 1134 what is left of time, not an amount of time, but the corpus does not by itself settle the lexical classification of *rest*.

1135 Under the semantic analysis, this subgroup has no pseudo-partitive use, and the earlier × in Table 1's pseudo-partitive column was substantively
 1136 right. The classification of *rest* is therefore left open rather than treated as a settled negative. What was wrong in the earlier draft was its justification: it
 1137 starred *the rest of time* as ungrammatical. The string is fine, and common.

Bare complement (the X of N). Hand-picked complements returned *the rest of time* 75; *the rest of people* 12; *the rest of day* 7; *the remainder of time* 3; *the remainder of day* 2; *the rest of money* 1, total 100. A wildcard sweep returns far more: *the rest of* [nn*] gives 2,820 tokens across 696 distinct bare complements and *the remainder of* [nn*] 126 across 96. The leading complements are *society* 442, *humanity* 248, *life* 150, *nature* 78, *mankind* 77, *time* 75, *eternity* 73, *world* 63, *creation* 42, *Americans* 41, *government* 31, *baseball* 26, *civilization* 26, *Congress* 21.

Those totals are not a partitive count, and the sweep has to be sorted by what the complement denotes before it says anything. Two classes fall out.

Where the complement denotes a BOUNDED UNIQUE WHOLE, the partitive reading is available and the determiner is unnecessary. Generic collectives are the productive case: *society* 442, *humanity* 248, *nature* 78, *mankind* 77, *creation* 42, *Americans* 41, *government* 31, *baseball* 26, *civilization* 26, *Congress* 21, and *class* 15. Each denotes uniquely without an article and each has proper parts, so a complement subset is well defined: *scientists and the rest of society*.

Where the complement denotes something *unbounded*, it doesn't. Bare *of time* offers nothing to partition, so *the rest of time* is not a partitive with a bare complement; it is a distinct duration expression meaning "all time henceforth", and it is largely fixed (*for the rest of time, till the rest of time*). *Eternity* 73 patterns with it. On the genuine partitive reading the determiner is obligatory, so **the remainder of time* stands as a starred datum once the reading is fixed. *Life* 150 is ambiguous between the two classes ("the rest of one's life", duration; "the rest of life on earth", collective) and isn't counted for either.

That resolves why the four-complement probe looked categorical. *Day, time, people, and money* contains no bounded generic collective at all, so it sampled only the class where the bare complement fails, and the 100 tokens it returned are duration uses, possessives, and modifier noise rather than partitives. The frame's productive class was outside the sample.

One caveat on the query. Because [nn*] matches a string position, some hits are the first word of a longer nominal (*the rest of day one, the rest of tuition/books/housing*), and *the rest of US* is mostly the country rather than the pronoun. The head of the distribution is unaffected.

Determined complement (the X of the N): *the rest of the day* 1,932; *the rest of the time* 830; *the rest of the people* 272; *the rest of the money* 222; *the remainder of the day* 75; *the remainder of the time* 21; *the remainder of the money* 13. Total: 3,365. These are genuine partitives on both criteria.

Indefinite-NP complement: *the remainder of a day* 1.

Determined-complement share on matched complements: $3,365 / (3,365 + 100 + 1) \approx 97.1\%$ raw. This figure is computable, because numerator and denominator draw on the same four complements, but note what it does and doesn't compare. It is not a partitive-versus-pseudo-partitive ratio: the classification of *rest* remains open, while *remainder* is excluded under the semantic analysis. It measures how often the whole carries a determiner, within one construction, for *day, time, people, and money*. For those four the determiner is strongly preferred, and that says nothing about the several hundred bare complements the sweep found.

A.3 OVERRIDE-DIRECTION AGREEMENT

Test of the "near-obligatory in central cases" claim. Subject NP plus finite verb agreement, by complement number/count.

Subject	Predicted	Plural agr.	Singular agr.	% predicted
<i>a lot of people</i>	plural	4,195 (ARE 3,406; WERE 789)	85 (IS 72; WAS 13)	98.0%
<i>a lot of money</i>	singular	1 (ARE, filtered)	90 (IS 61; WAS 29)	98.9%
<i>a number of people</i>	plural	100 (ARE 52; WERE 48)	0 (4 false positives)	100%
<i>lots of people</i>	plural	348 (ARE 288; WERE 60)	0 (6 false positives/non-standard)	100%
<i>lots of money</i>	singular	0 (1 false positive)	24 (IS 19; WAS 5)	100%
<i>plenty of people</i>	plural	79 (ARE 65; WERE 14)	0	100%
<i>plenty of money</i>	singular	0	6 (IS 4; WAS 2)	100%
<i>the rest of the people</i>	plural	19 (ARE 15; WERE 4)	0	100%
<i>the rest of the money</i>	singular	0	19 (IS 10; WAS 9)	100%

Override is in the predicted direction at 98–100% across all tested QN+complement combinations after filtering. Of these, the discriminating cells, where the head's number and the complement's diverge (*a lot of people, a number of people, lots of money, the rest of the people*), are what evidence the override; the matched cells (*a lot of money, lots of people, the rest of the money*) confirm the predicted direction but are equally consistent with head-driven agreement, so they don't discriminate. The filtering below was applied to counter-direction cells; predicted-direction cells were not hand-scrubbed to the same depth, so these figures are exploratory corroboration of the categorical override, not estimated rates. *A lot of money* required filtering: of 13 raw plural-agreement hits, 10 were parse-shifted (the surface string *a lot of money are* appearing inside relative clauses modifying a plural-noun head, e.g., *people who earn a lot of money are successful; movies that make a lot of money are the biggest help; times when we... make a lot of money are gone*), 1 was the irrealis *were* (*if a lot of money were at stake*), and 1 's hit was singular *has* (matches prediction); only 1 hit was a genuine plural override (*a lot of money are being raised*). Filtered count: 1 plural / 90 singular = 98.9% singular.

The other checked counter-direction cells are artefacts. *A number of people is/was* is headed by *condition, suggestion, ability, or counsel*, not by *number*; *lots of people is/was* is headed by expressions such as *murder, hauling, or disappearance*, with one non-standard existential hit; and *lots of money are* occurs inside a PP modifying plural *men*. The large *a lot of people is/was* cell remains unfiltered, so the 98.0% row is conservative.

A.4 BUNCH: ANIMATE VS. INANIMATE AGGREGATES

Testing CGEL's claim (§18.2, p. 503) that *bunch* tilts collective for inanimate aggregates and transparent for groups of people. Animate *bunch*-subjects are consistently plural across *a bunch of people* and *a bunch of kids*, supporting the transparent half. The inanimate half gets no positive support: inanimate *bunch* is rare in subject position, its typical use being as object; CGEL's own example *A bunch of flowers was presented to the teacher* returns no hits; and the few inanimate-subject tokens take plural rather than the predicted singular. Counts are small enough that this bears only on whether the contrast projects as a corpus regularity, not on the grammatical claim, and the full returns are in the supplementary material.

1186 A.5 NEGATIVE CONTROL: REFERENTIAL *THE NUMBER OF*

1187 If the override tracked the quantificational construction rather than the noun *number* or *N-of-N* configurations in general, then referential *the number*
 1188 *of*, where *number* is the head rather than a quantifier, should agree head-driven (singular) even with a plural complement. A COCA check (June 2026)
 1189 bears this out on the same lexeme. *The number of people* occurs with *is/are* at 3/1 and *the number of cases* at 20/1: head-driven singular, with plural
 1190 only at attraction level. *A number of people* occurs at 0/52 and *a number of cases* at 0/3: categorical plural override. The counts are small, since the bare
 1191 subject-verb adjacency is rare, but the direction is unambiguous and the contrast sits on one noun: *number* overrides under *a* and not under *the*. The
 1192 override is therefore a property of the quantificational construction, not of the lexeme or of *N-of-N* structure as such.

1193 ACCESSIBILITY TOPOLOGIES BEYOND THE WORKED CASES

1194 Spanish and Tsez (§4.3) are the cross-linguistic cases the body of the paper works. This appendix sets out four further
 1195 accessibility topologies the schema maps, each with the falsification condition that would retire it, plus two notes on what
 1196 the table's entries do and don't establish. None is promoted to a worked projectibility profile; the point is to show the
 1197 schema's reach and its edges, and to mark which entries lack a concealing control. Every case remains subject to the §2.3
 1198 concealing-baseline test: identify *X*, *P*, and *R*, then ask what same-language pattern would make the proposed access
 1199 disappear.

1200 Table 2. Accessibility topologies under the umbrella schema.

Topology	<i>X-P-R</i> binding	Falsification condition and status
Outward endpoint access	<i>X</i> = NP or clause mediator; <i>P</i> = complement or embedded feature; <i>R</i> = external agreement. English QNs and transparent free relatives, Spanish partitives, and Tsez long-distance agreement instantiate this topology.	Fails where only the mediator's own head/class feature controls <i>R</i> , or where the construction-type has no concealing variant. This is the paper's core positive pattern.
Null-source subpart access	<i>X</i> = predicate configuration containing a possessed NP with an unpronounced possessor; <i>P</i> = the recoverable possessor's person/number; <i>R</i> = anaphoric verbal morphology plus subject case. Aleut is the key case (Merchant, 2011, p. 195).	Fails if the omitted possessor patterns like an ordinary overt possessor, or if the anaphoric verbal suffix appears without the missing subpart. Positive with a qualification: Merchant analyzes the missing possessor as syntactically present <i>pro</i> .
Inward dependent access	<i>X</i> = clause or nominal-dependency domain; <i>P</i> = clausal TAM/case value; <i>R</i> = morphology on nominal dependents. Lardil and Kayardild show clausal TAM values on dependent nominals (Nordlinger & Sadler, 2004).	Fails if dependent marking is only local nominal morphology, with no covariation with clausal predication. This is an observational topology; no generative analysis is needed for the comparison.
Path-distributed access	<i>X</i> = extraction path; <i>P</i> = grammatical relation/case of the extracted item; <i>R</i> = <i>wh</i> -agreement morphology along that path. Chamorro supplies the comparison (Chung, 1998, 2004).	Fails if the morphology appears only at the gap or endpoint, or if it doesn't vary with the extracted item's role. Positive as a topology, but extraction remains outside this paper's worked evidence.
Feature relay boundary case	<i>X</i> = possessed NP; <i>P</i> = possessor obviation/proximation; <i>R</i> = possessed-noun morphology and verbal obviation agreement. Kutenai and Algonquian obviation are useful near misses (Dryer, 1992).	Fails as transparency where the predicate sees only a feature already relayed onto the possessed head. Boundary case: transformed access, outside the direct-mediated pattern.

1201 The table is a topology map, not a second set of worked cases. Aleut is the qualified positive; the head-marked and
 1202 obviation entries remain conditional on the analyses and falsification conditions stated in the table. Lardil, Kayardild,

1203 and Chamorro mark candidate applications rather than established instances because an independently motivated same-
1204 family concealing control has not been shown.

1205 DATA AVAILABILITY

1206 The corpus material behind §2 and Appendix A.3 is the COCA and COHA pilot described there: the full query list, raw
1207 returns, per-cell coding decisions, and the pre-registered protocol and results for the *piles* candidate-member test. These
1208 files travel with the paper as supplementary material and are versioned in the project repository at [https://github.com/](https://github.com/BrettRey/Linguistic_transparency)
1209 [BrettRey/Linguistic_transparency](https://github.com/BrettRey/Linguistic_transparency). A citable archive will be deposited on acceptance; the supplementary README
1210 records the manuscript commit so that every reported count can be traced to its query.

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